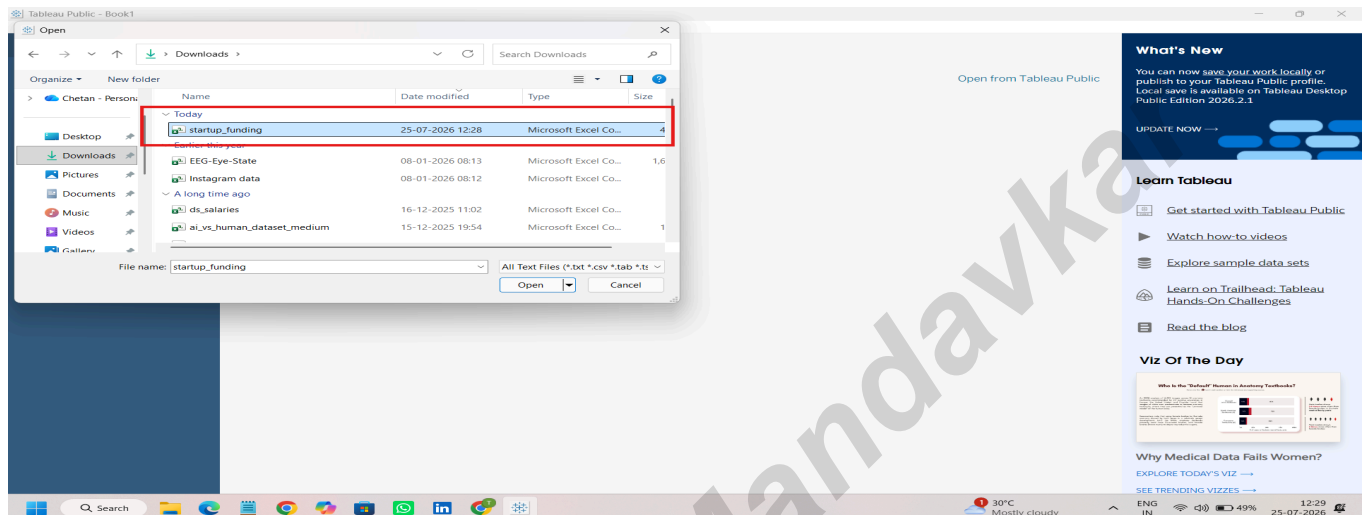


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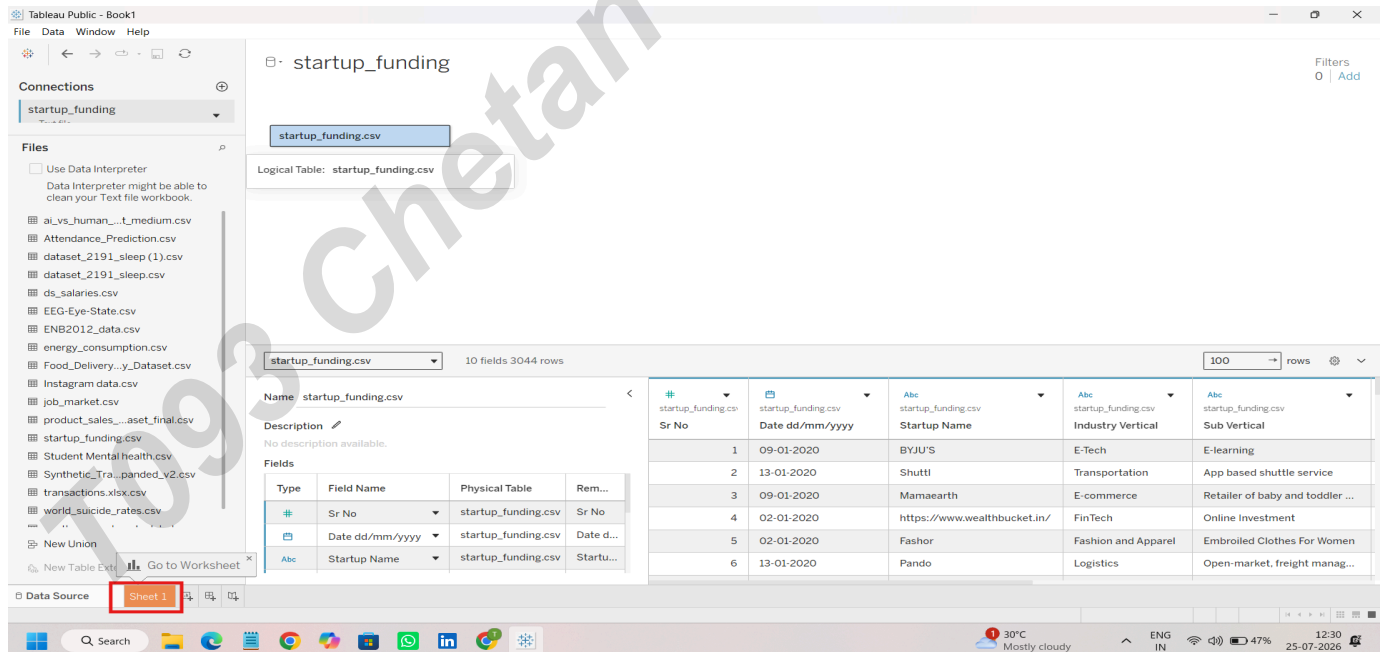
Aim:- Creating Advanced Visualizations in Tableau

A] Creating Stacked and Grouped Bar Charts for Comparison

Step 1:- Open Tableau And Connect The Dataset



Step 2:- Click on Sheet 1 To Enter In Workspace



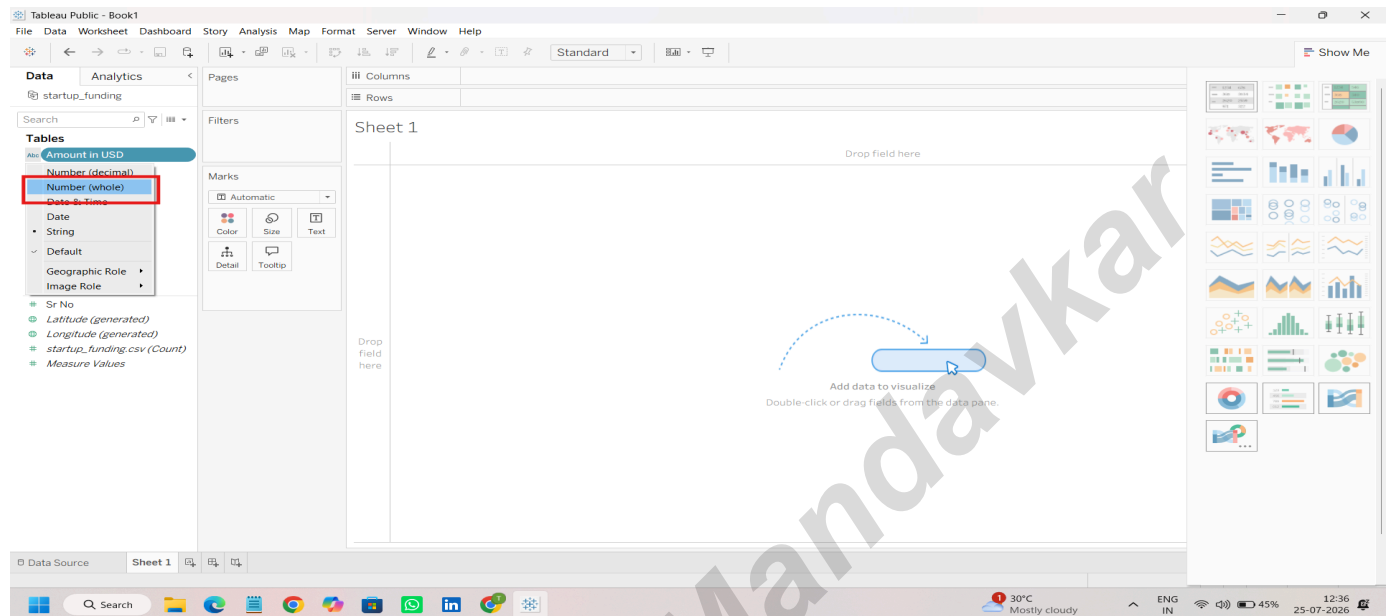
Name:- Chetan Mandavkar

Roll No. T093

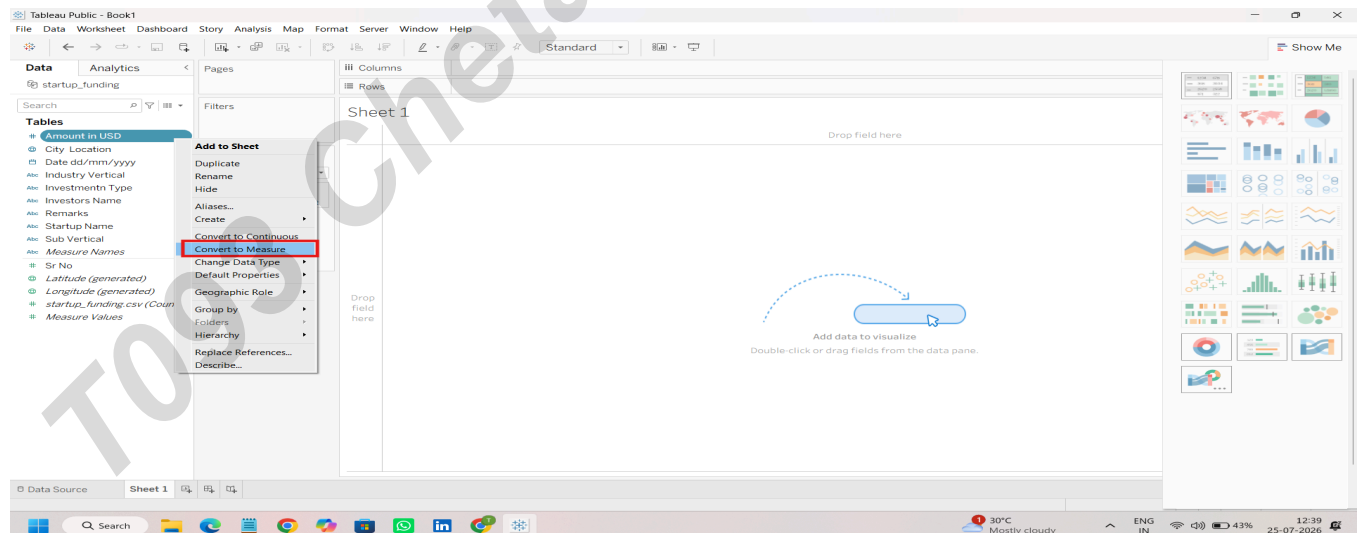
Subject:- Data Visualization With PowerBI And Tableau

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Step 3:- In Data Pane Which Is Located In Left Side, There is Amount in USD Convert This Field Into Number(whole) By Clicking small abc icon



Step 4:- Right Click On Amount In USD Amount And Select The Option Convert The Measure It will turn green and move to the bottom half of the pane.

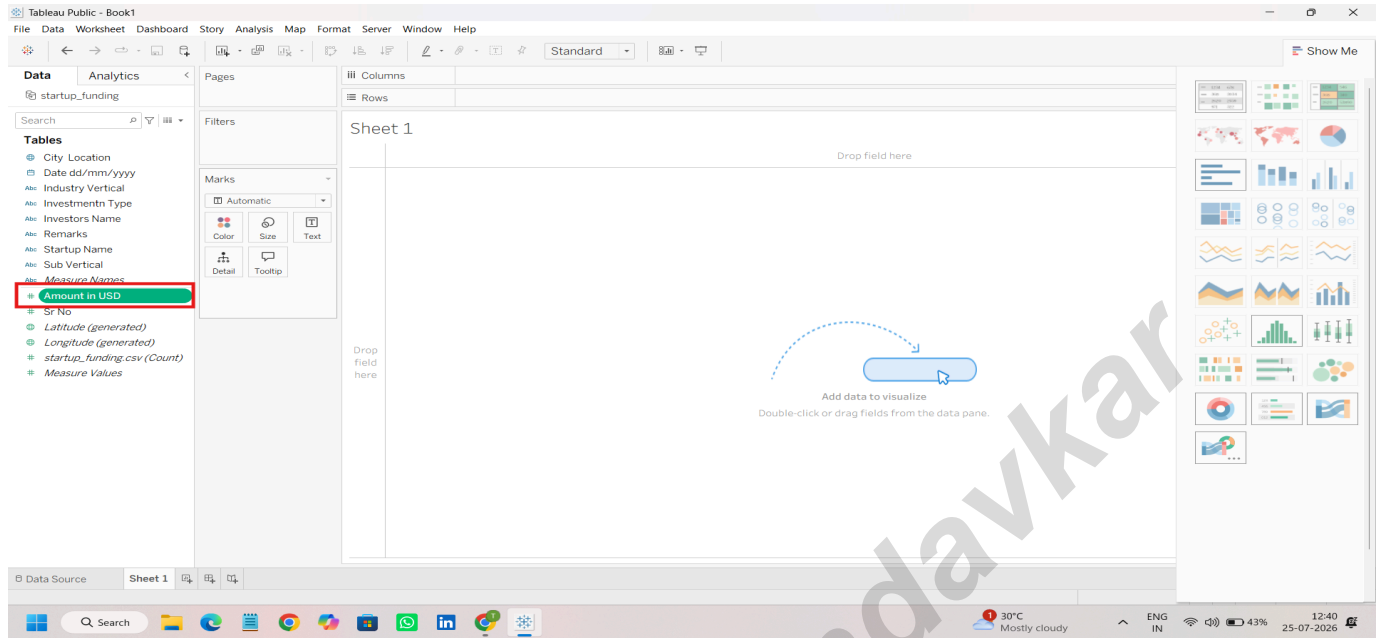


Name:- Chetan Mandavkar

Roll No. T093

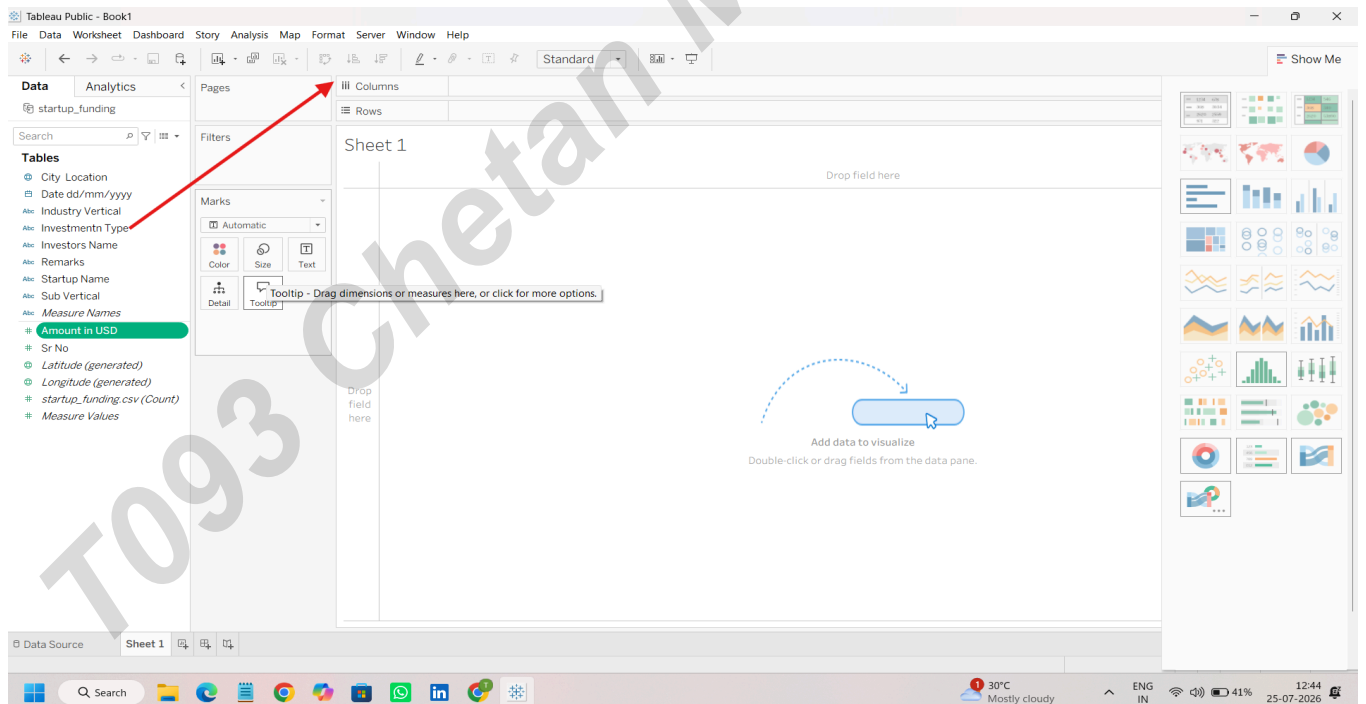
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Step 5:- Build The Stacked Bar Chart

a) From The Data Pane Drag The InvestmentnType Field Into Columns

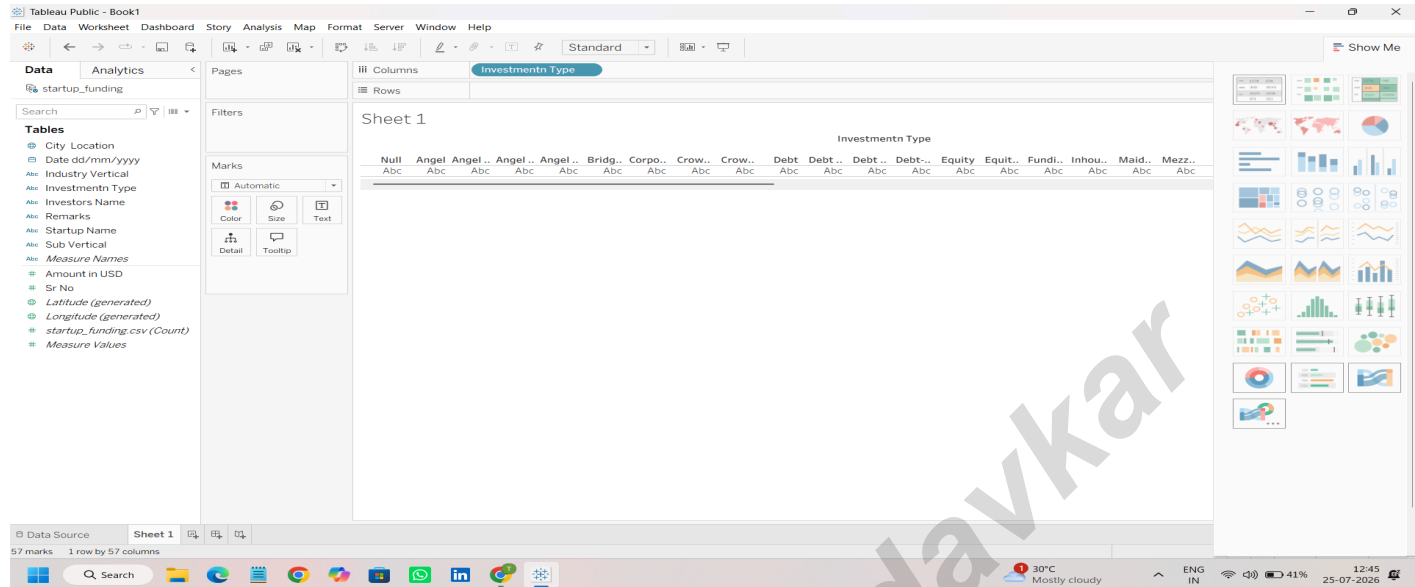


Name:- Chetan Mandavkar

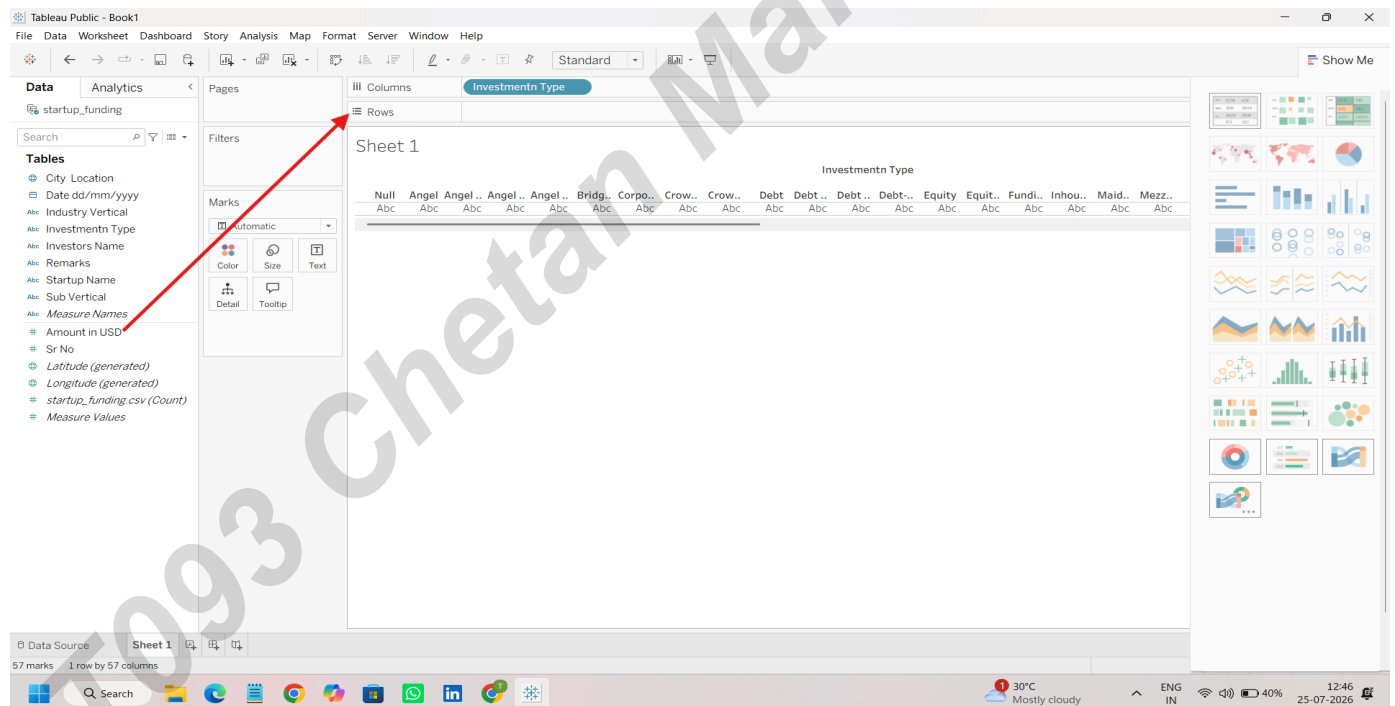
Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

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b) From The Data Pane Drag The Amount In USD Into Rows

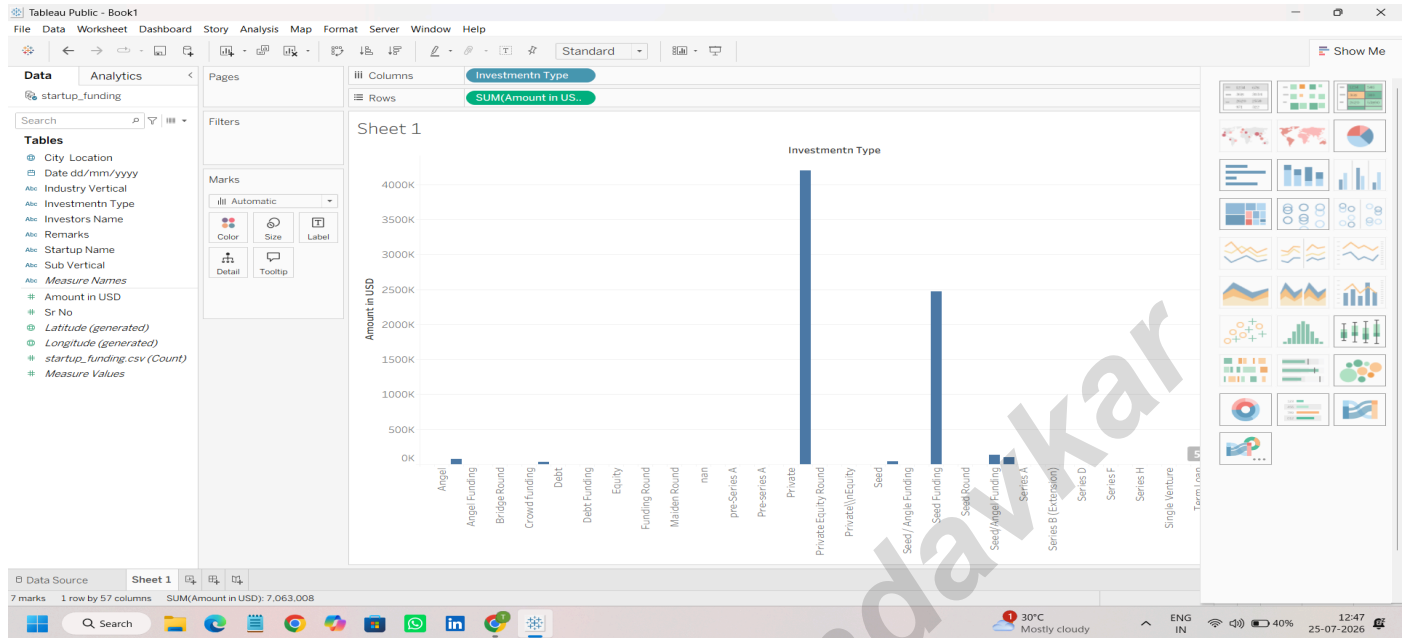


Name:- Chetan Mandavkar

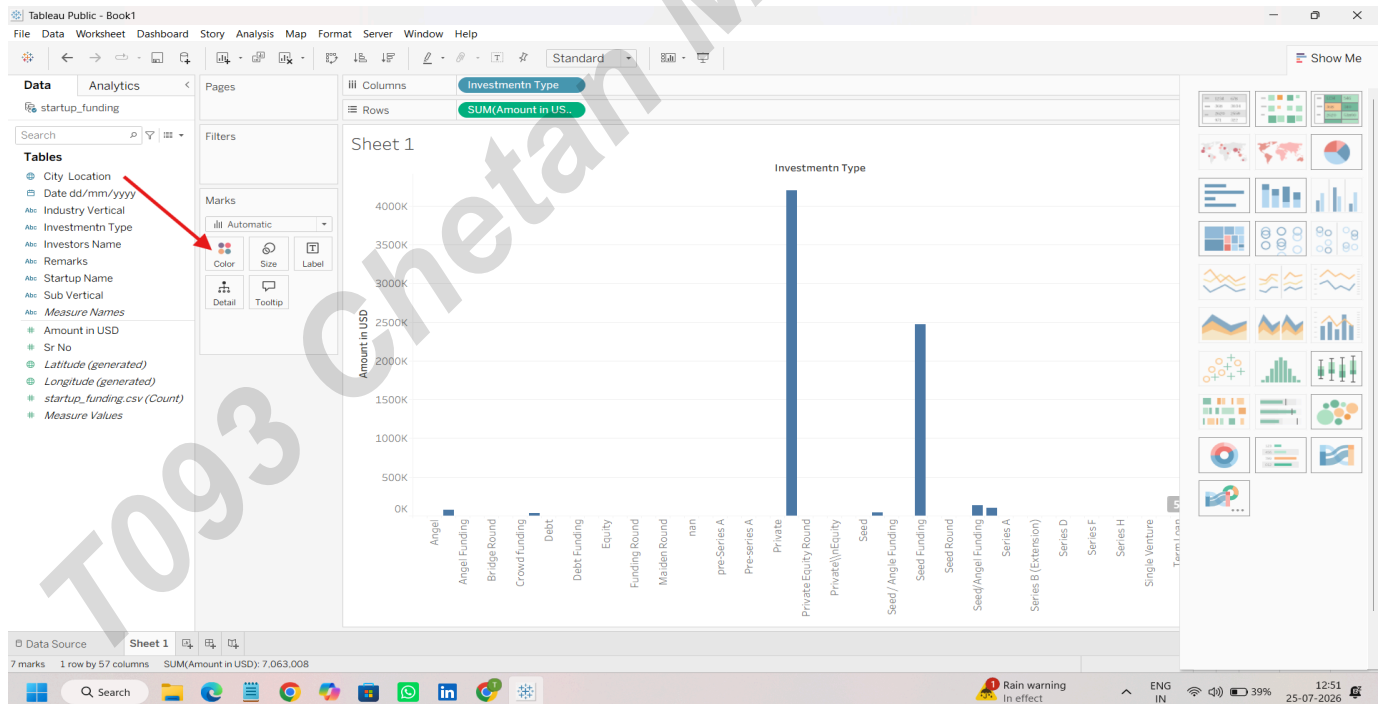
Roll No. T093

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c] Take The City Location Field From Data Pane And Drag into Directly Color Tile Which Is In Mark Card

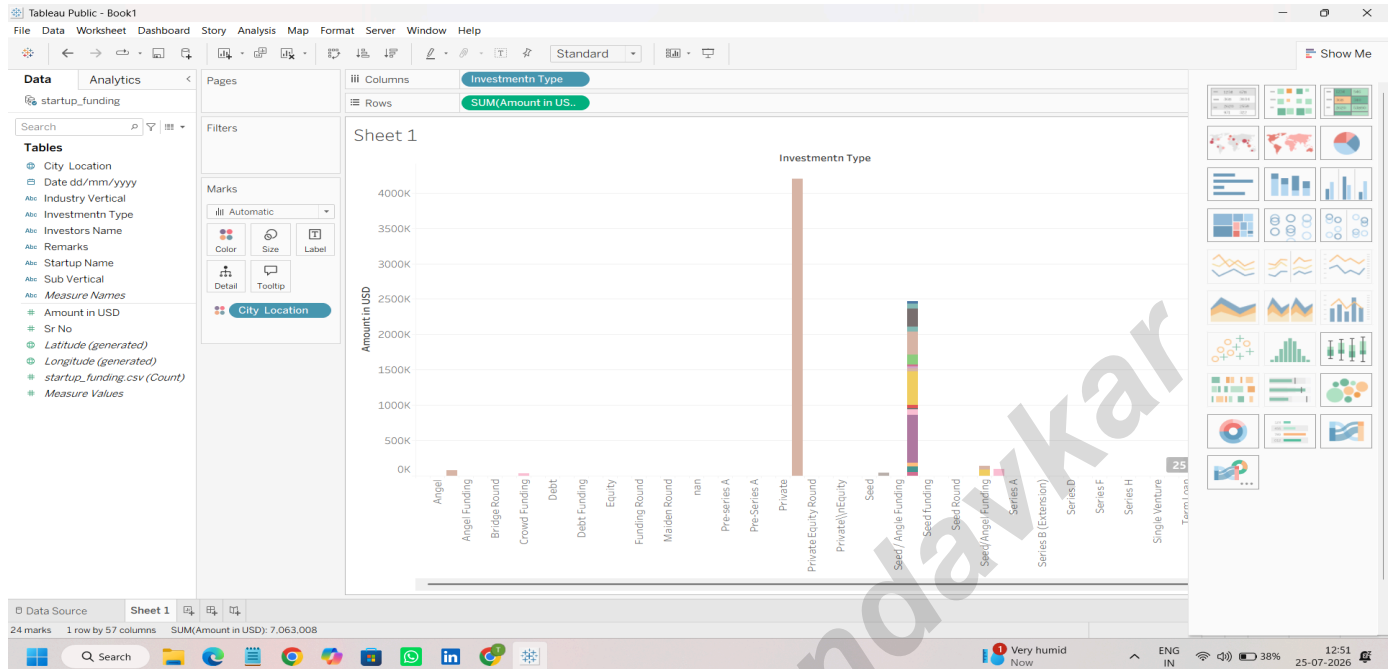


Name:- Chetan Mandavkar

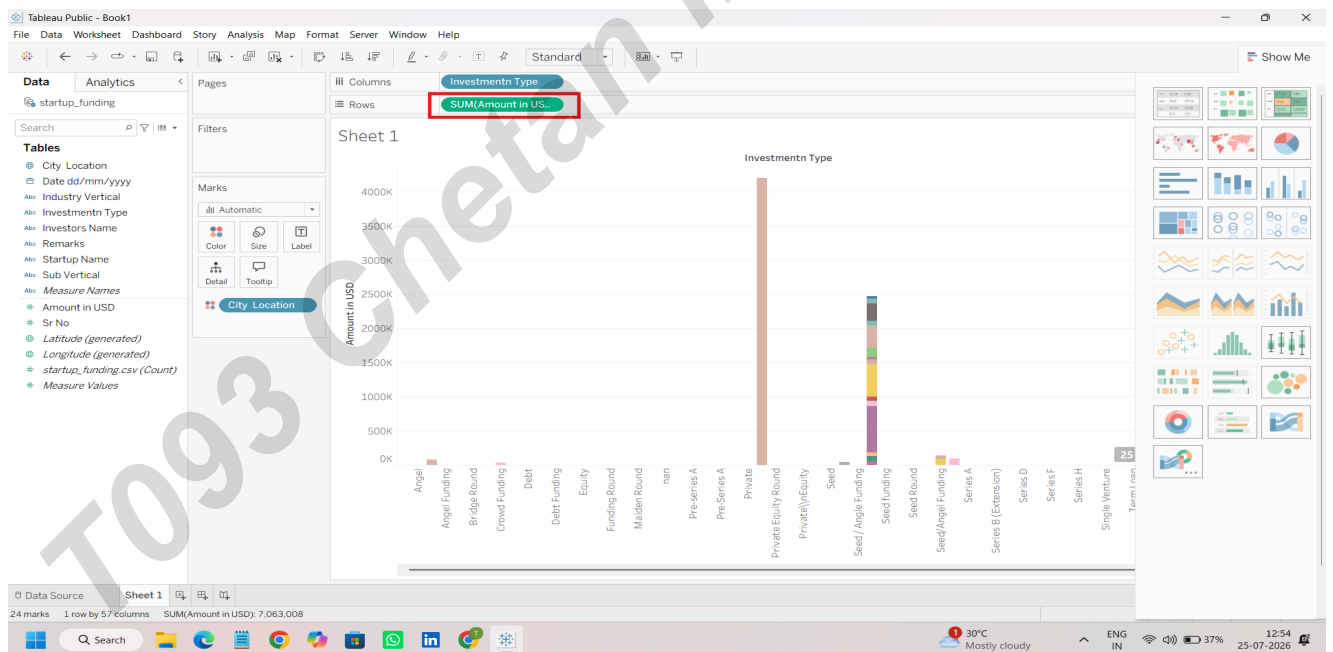
Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

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Step 6:- Converting To Grouped Bar Chart
a] Leave Amount in USD on the Rows shelf.

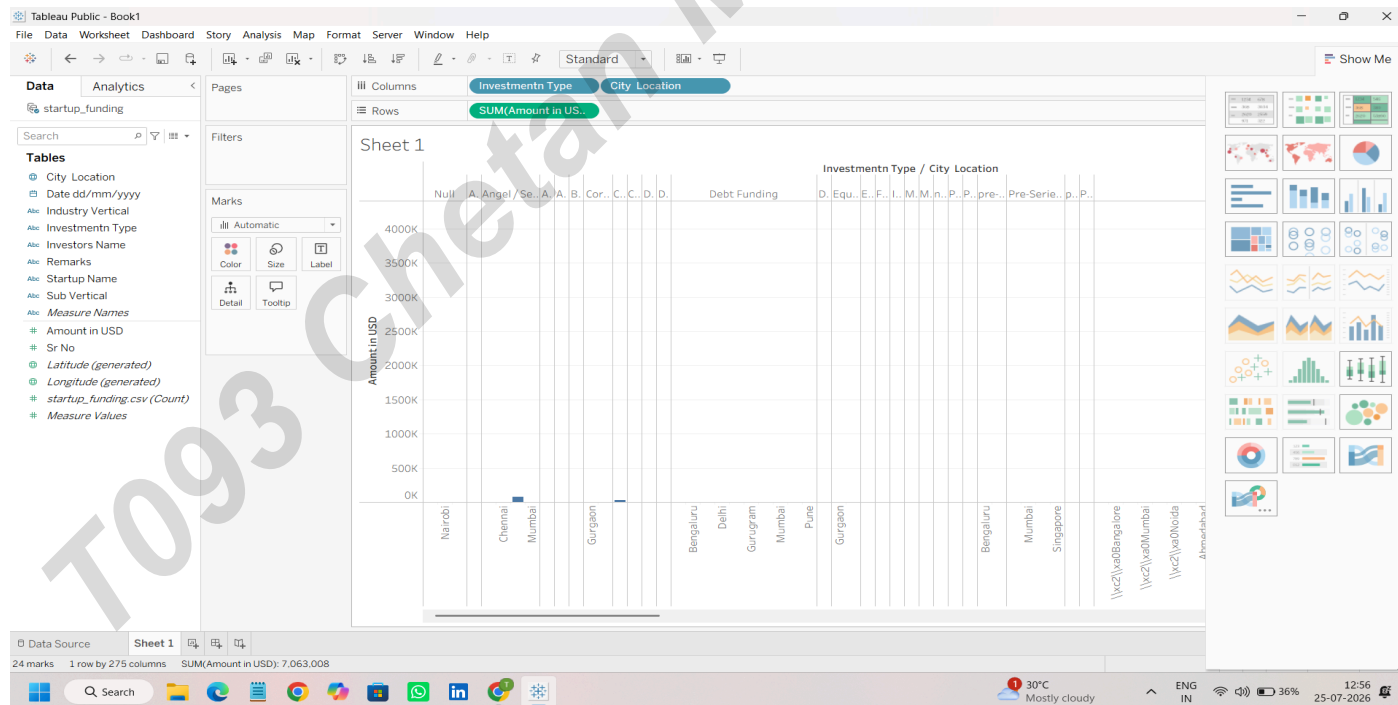


Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

b] Take the City Location pill out of the Color tile on the Marks card and drag it up to the Columns shelf, placing it right next to (to the left of) the InvestmentnType pill.

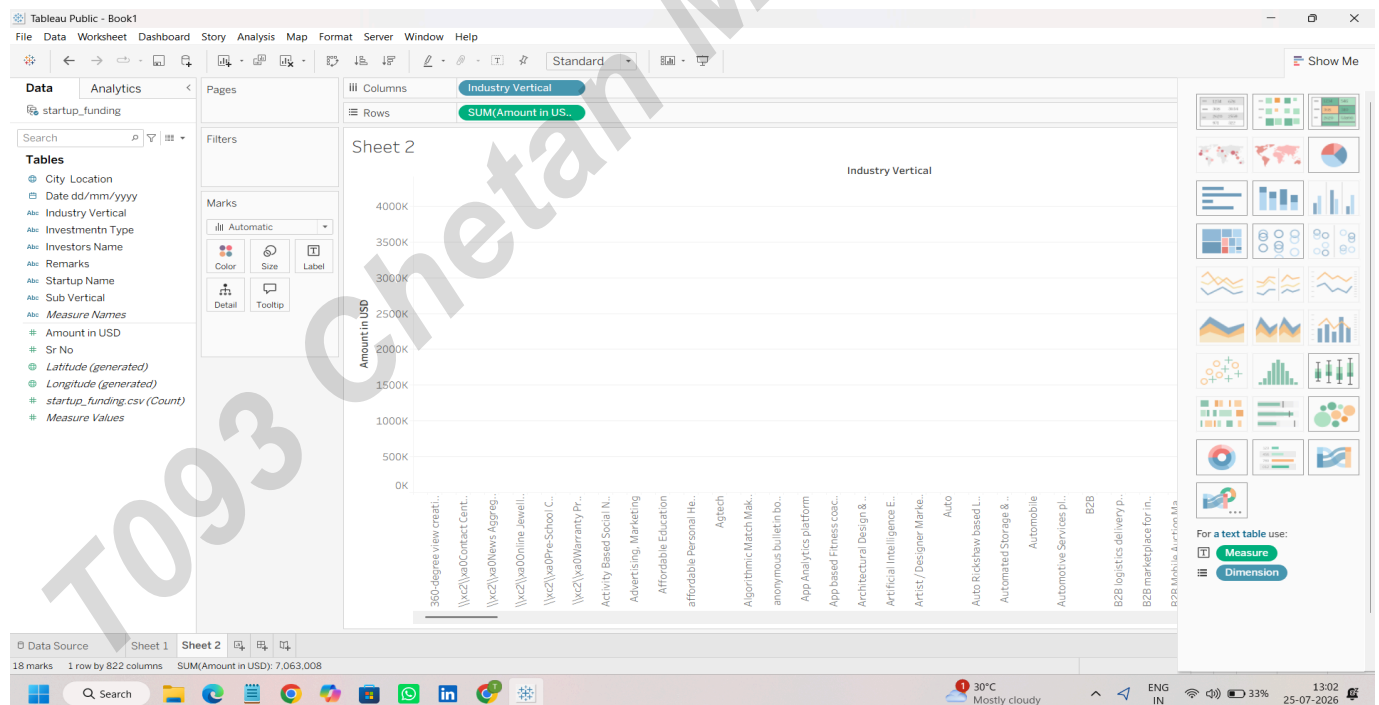
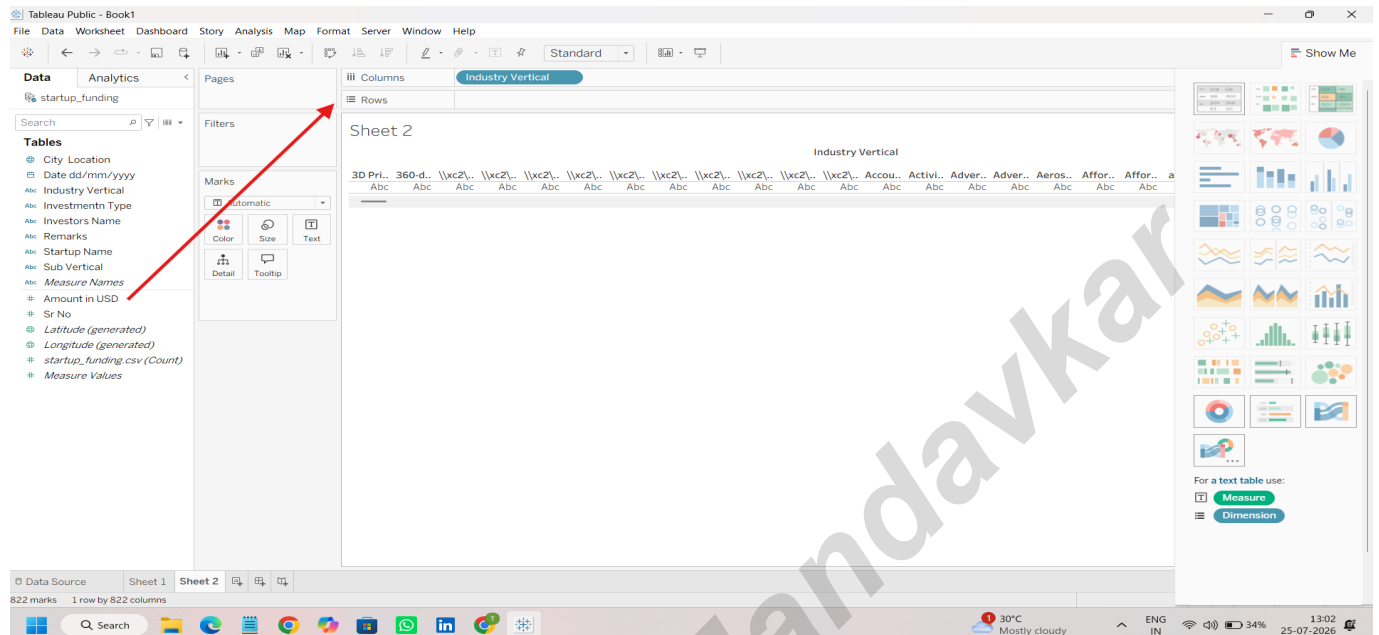


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Step 2:- Drag the Amount in USD measure and drop it onto the Rows shelf.



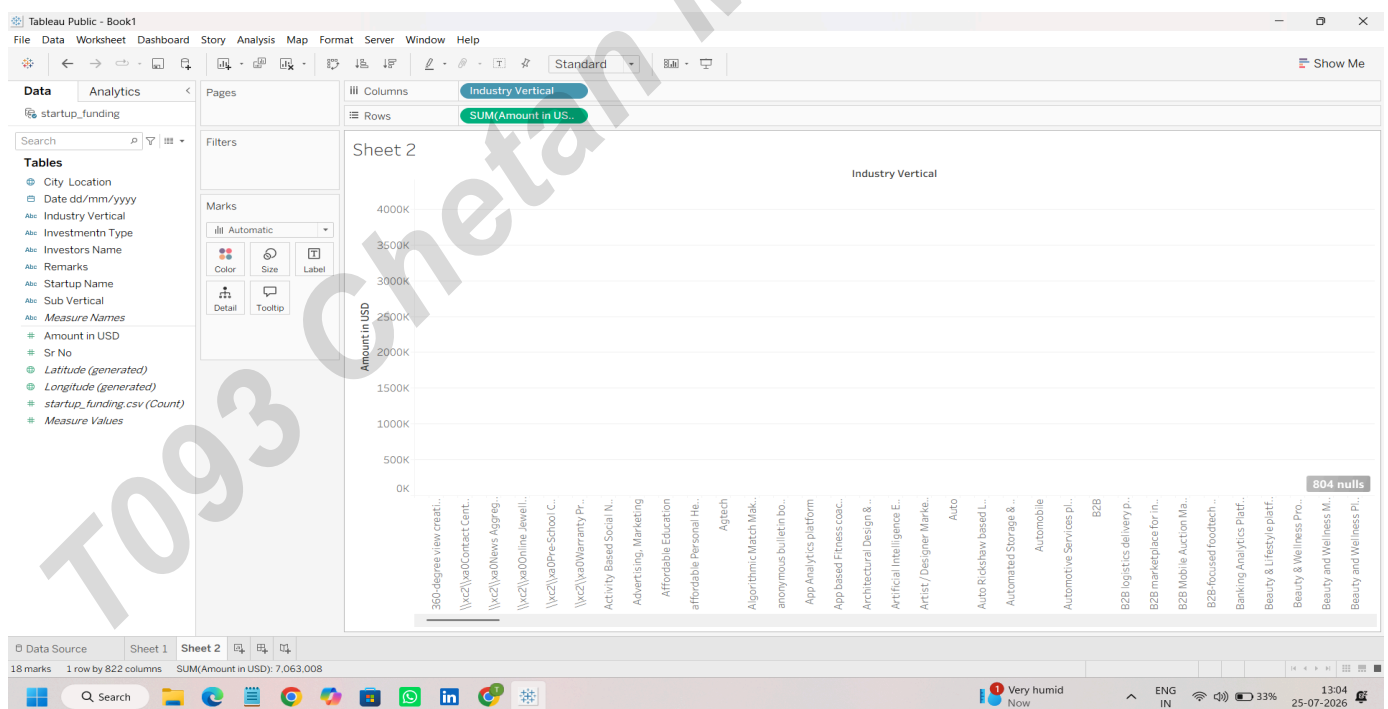
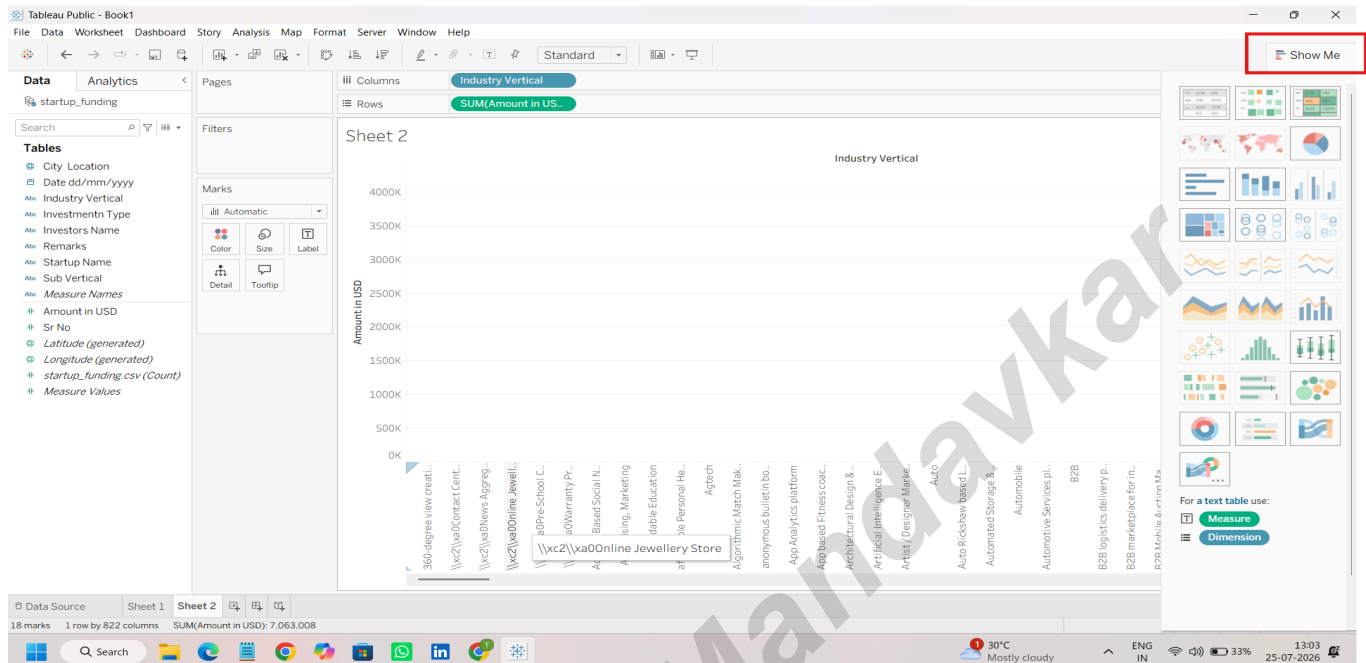
Name:- Chetan Mandavkar

Roll No. T093

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Step 3:- Now, look at the very top-right corner of the window and click the **Show Me** button to expand the chart library panel.



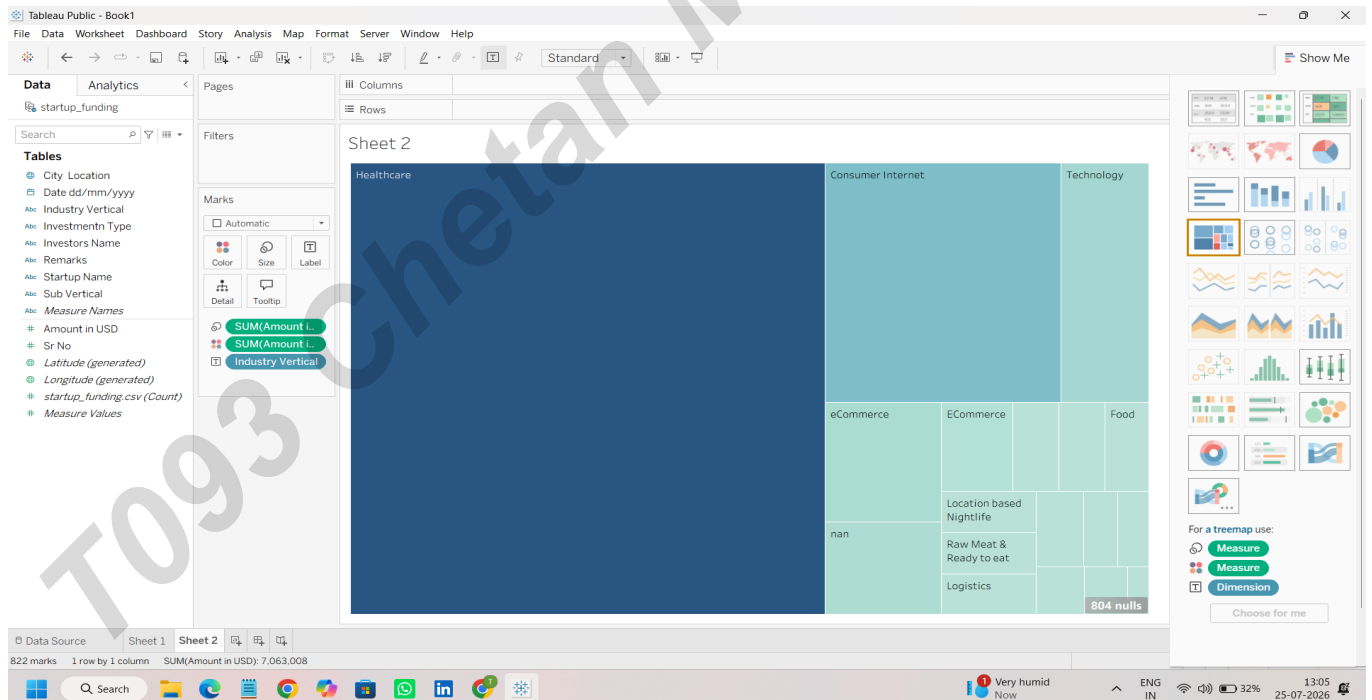
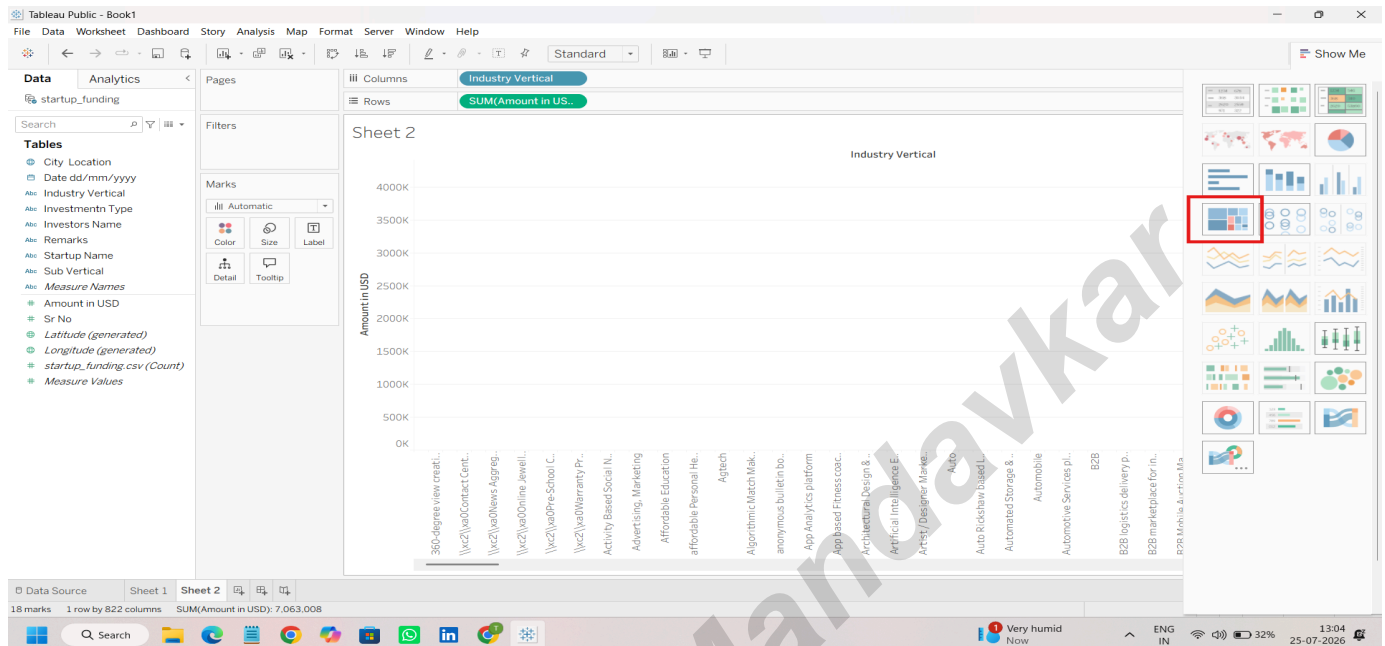
Name:- Chetan Mandavkar

Roll No. T093

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Step 4:- From the options, click on the Treemap icon (it looks like a grid of nested colorful rectangles).



Name:- Chetan Mandavkar

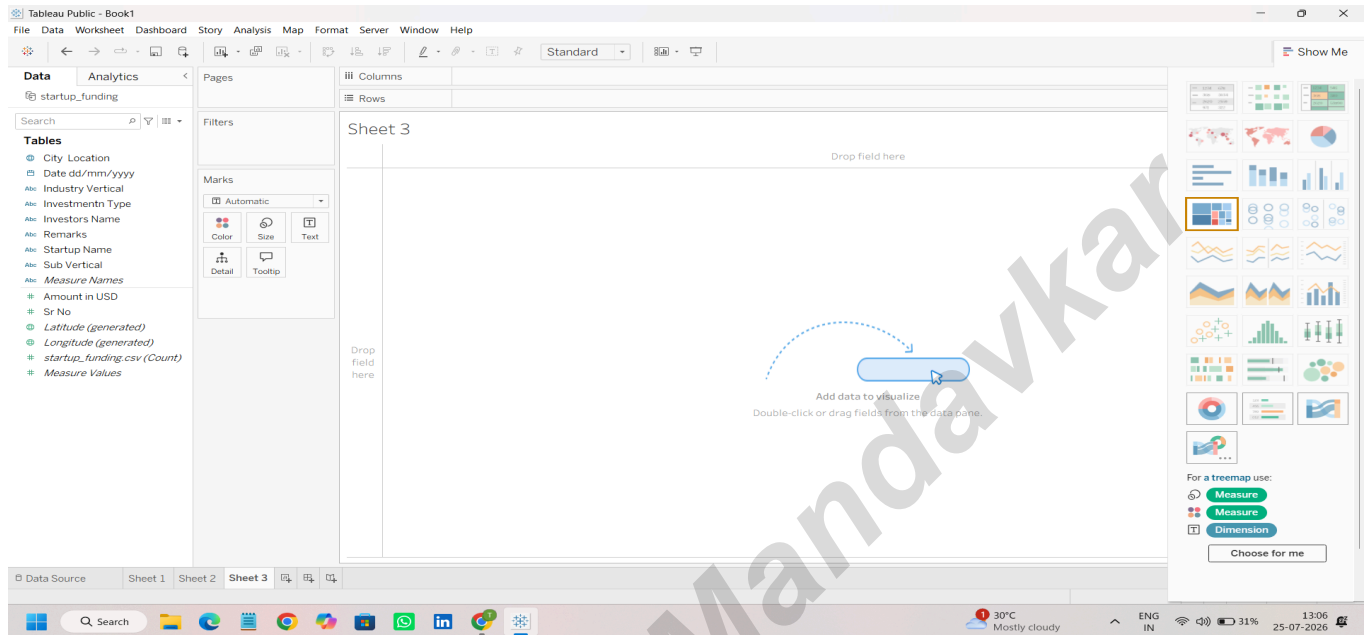
Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

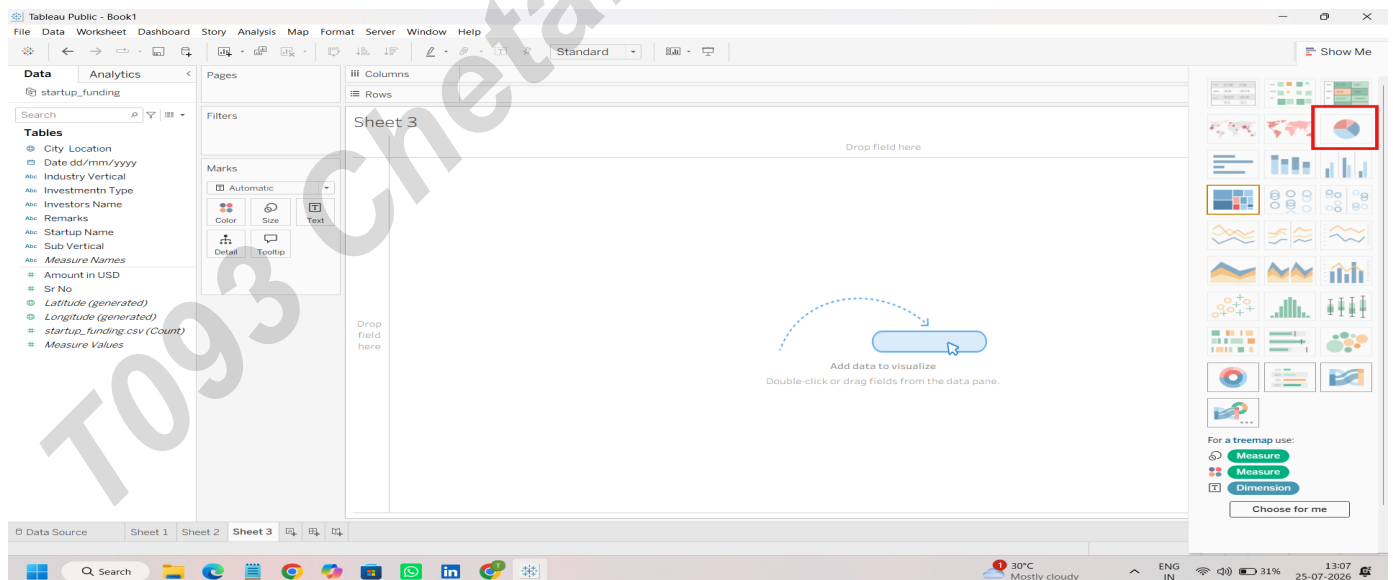
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Step 5:- Building A Pie Chart

a] Click the New Worksheet icon to open another blank layout (Sheet 3).



b] Go back to the Show Me panel in the top-right corner and select the Pie Chart icon.



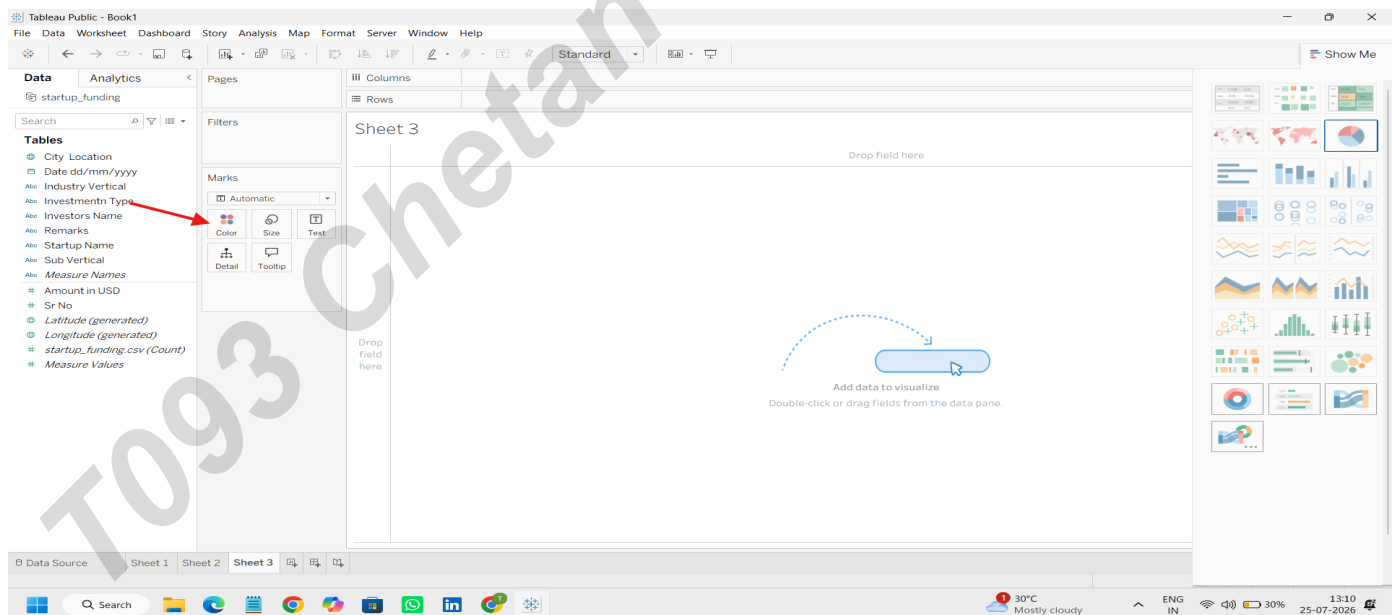
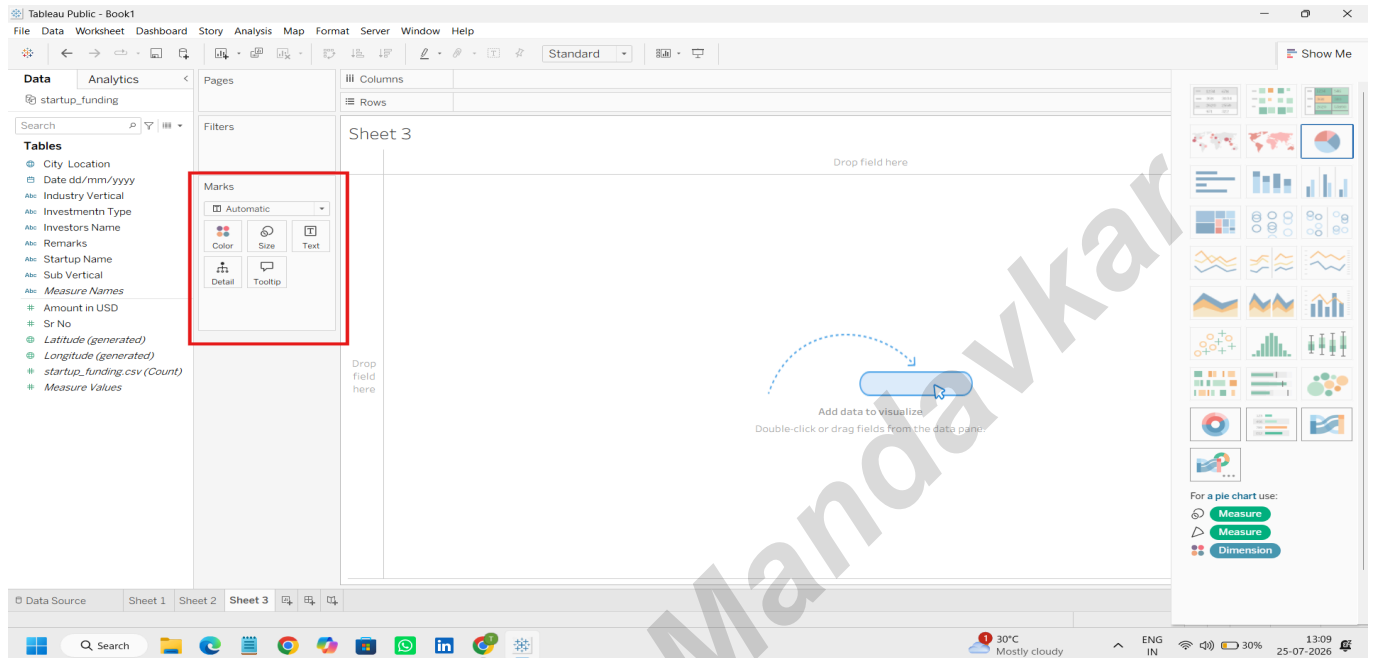
Name:- Chetan Mandavkar

Roll No. T093

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c] Notice that your Marks card changes to show an angle option. From the Data Pane, drag InvestmentnType and drop it onto the Color tile in the Marks card.

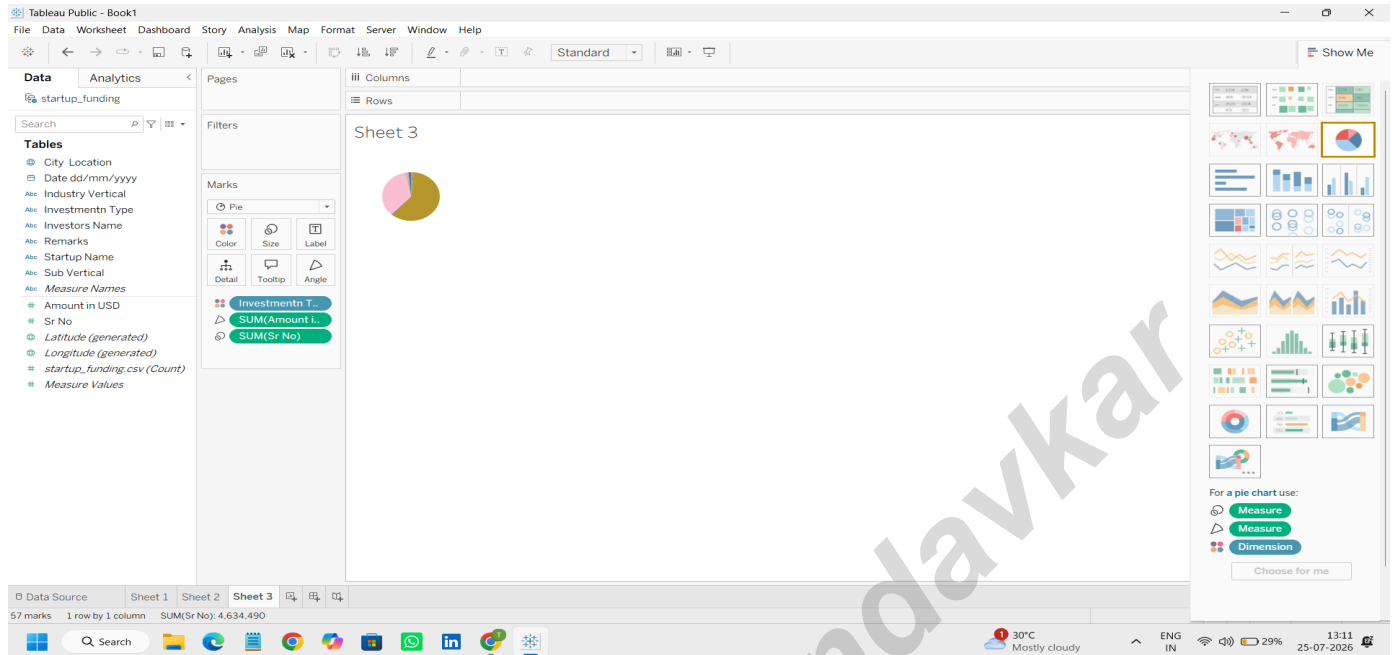


Name:- Chetan Mandavkar

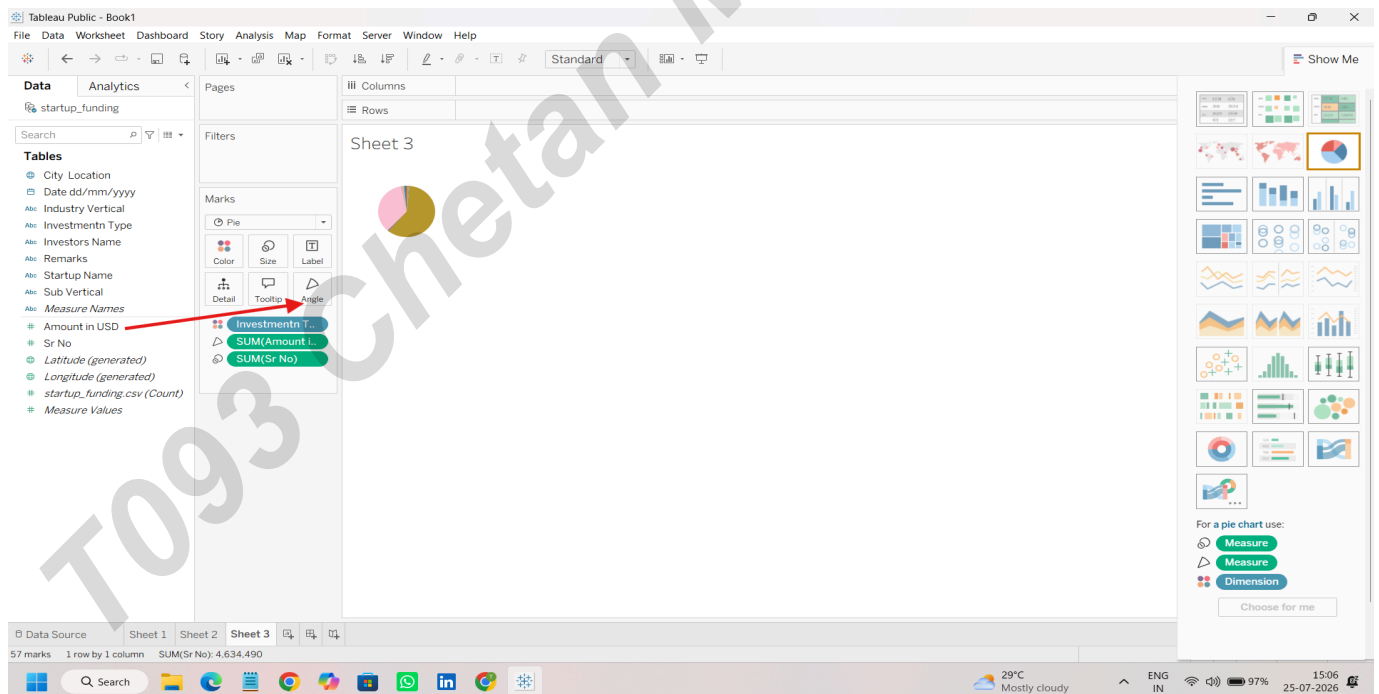
Roll No. T093

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d] Drag Amount in USD from the Data Pane and drop it directly onto the Angle tile in the Marks card.

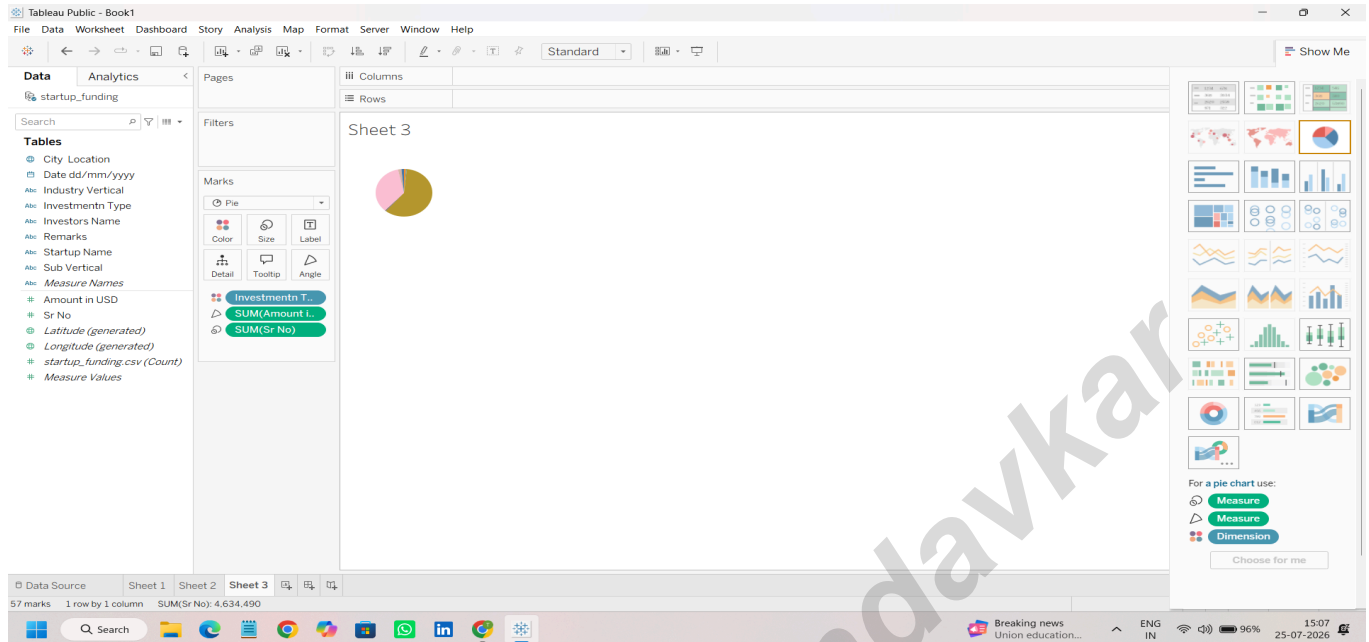


Name:- Chetan Mandavkar

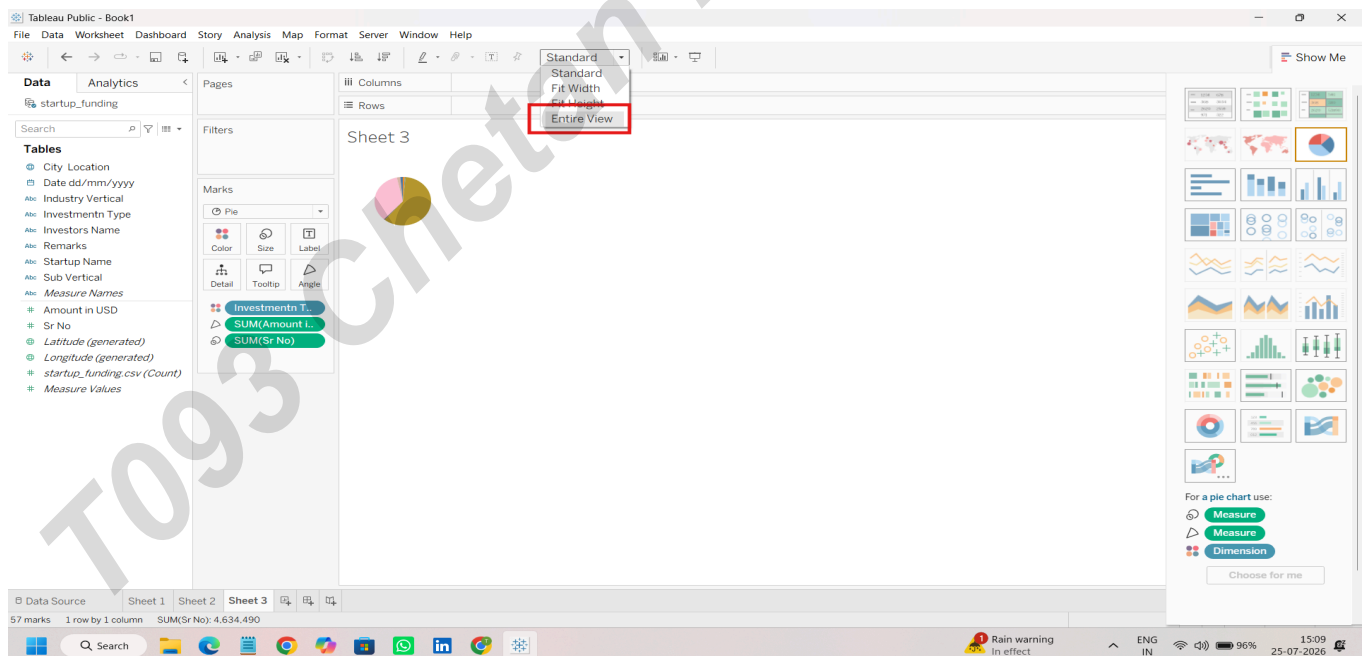
Roll No. T093

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e] Enlarge the view: To make the pie chart legible, look at the top toolbar menu, find the fit dropdown menu (which says Standard by default), and switch it to Entire View.

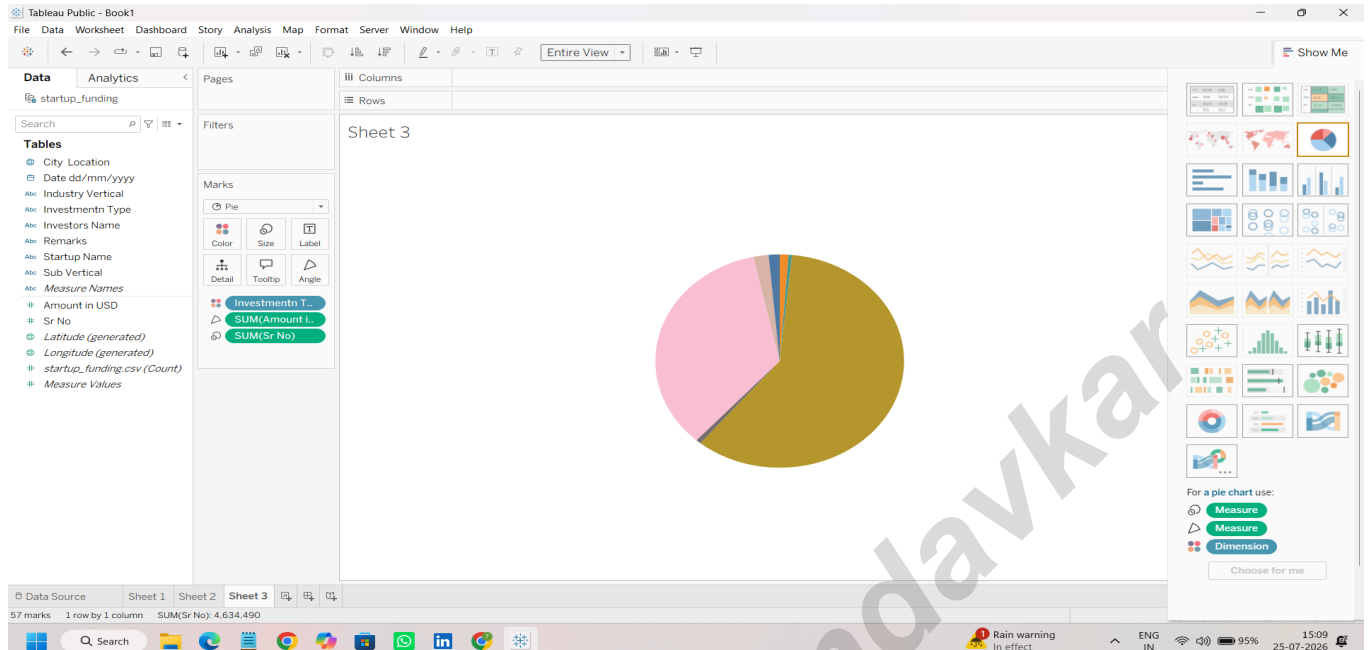


Name:- Chetan Mandavkar

Roll No. T093

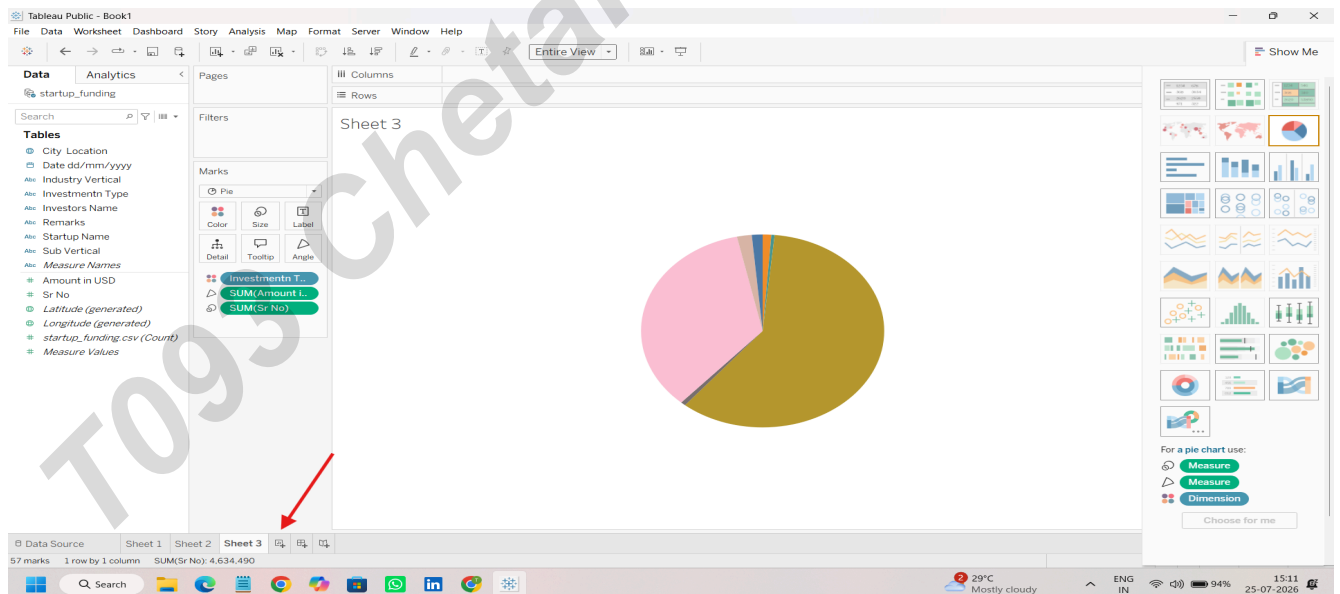
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C] Design scatter plots to identify relationships

Step 1:- Click the New Worksheet icon at the bottom of your window to open a fresh canvas (Sheet 4).



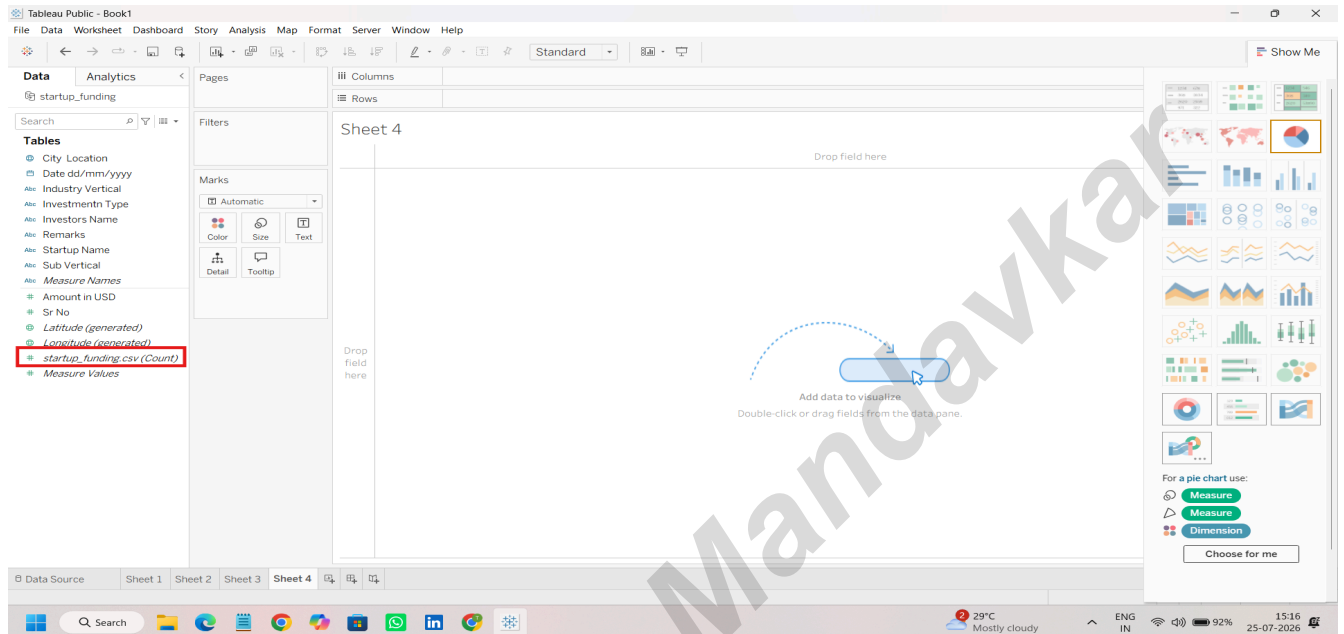
Name:- Chetan Mandavkar

Roll No. T093

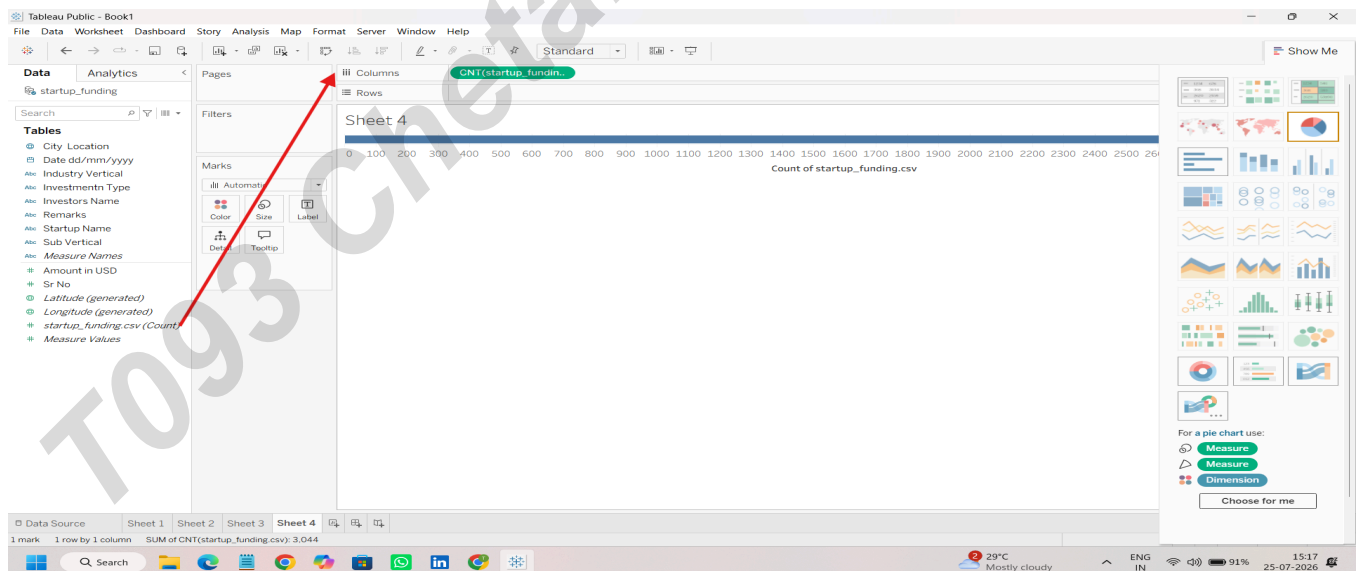
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Step 2:- Look down to the bottom of your Data Pane list under the green measures section. Locate the automatic system measure field called `startup_funding.csv (Count)` (in older versions of Tableau, this is called Number of Records).



Step 3:- Drag The Count Field Into Column Shelf



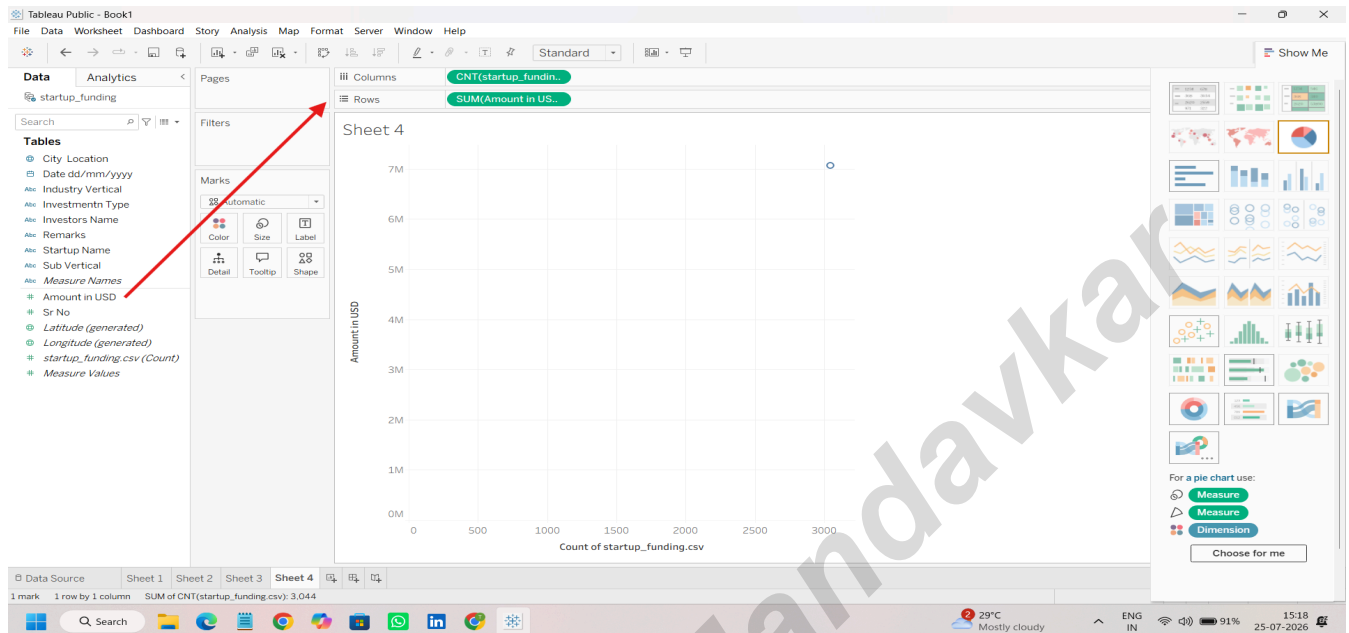
Name:- Chetan Mandavkar

Roll No. T093

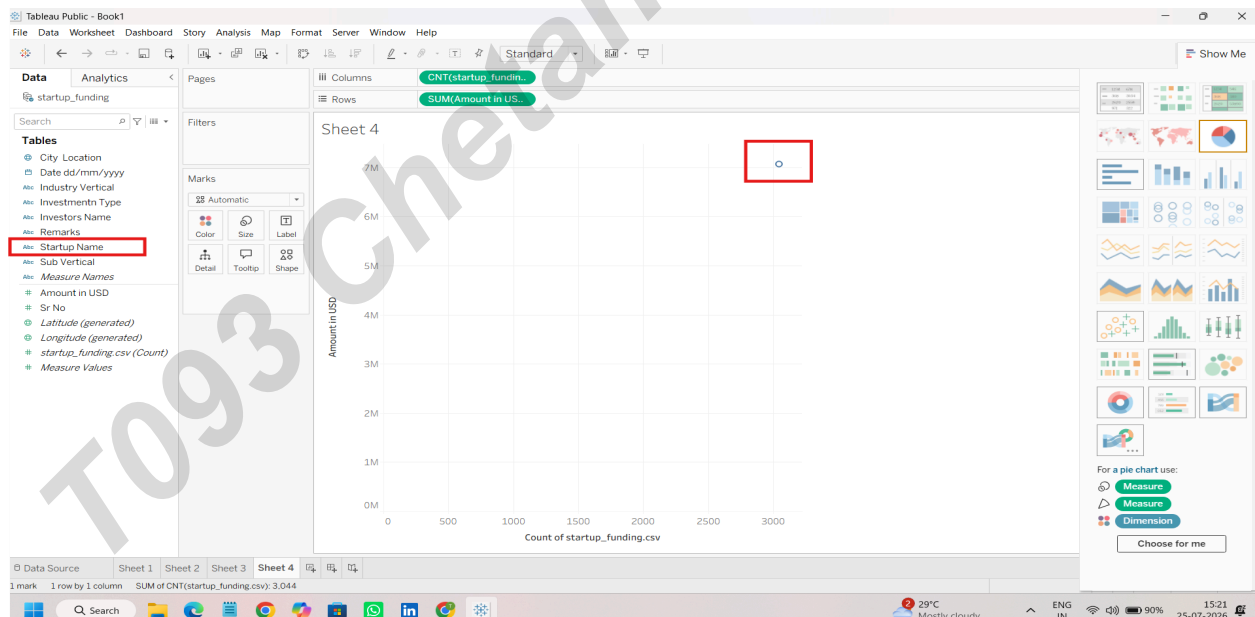
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Step 4:- Drag your Amount in USD measure and drop it onto the Rows shelf.



Step 5:- Now Break this single point down into individual startups, find the Startup Name dimension in the Data Pane.



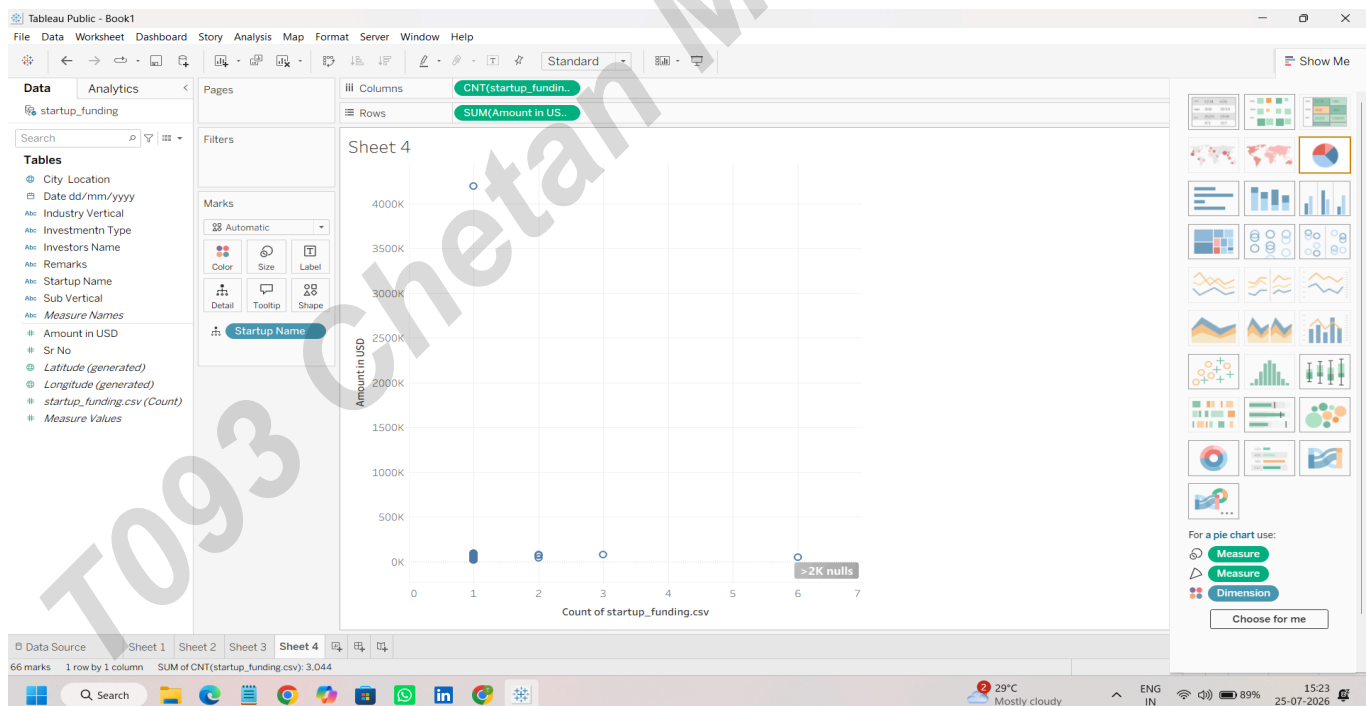
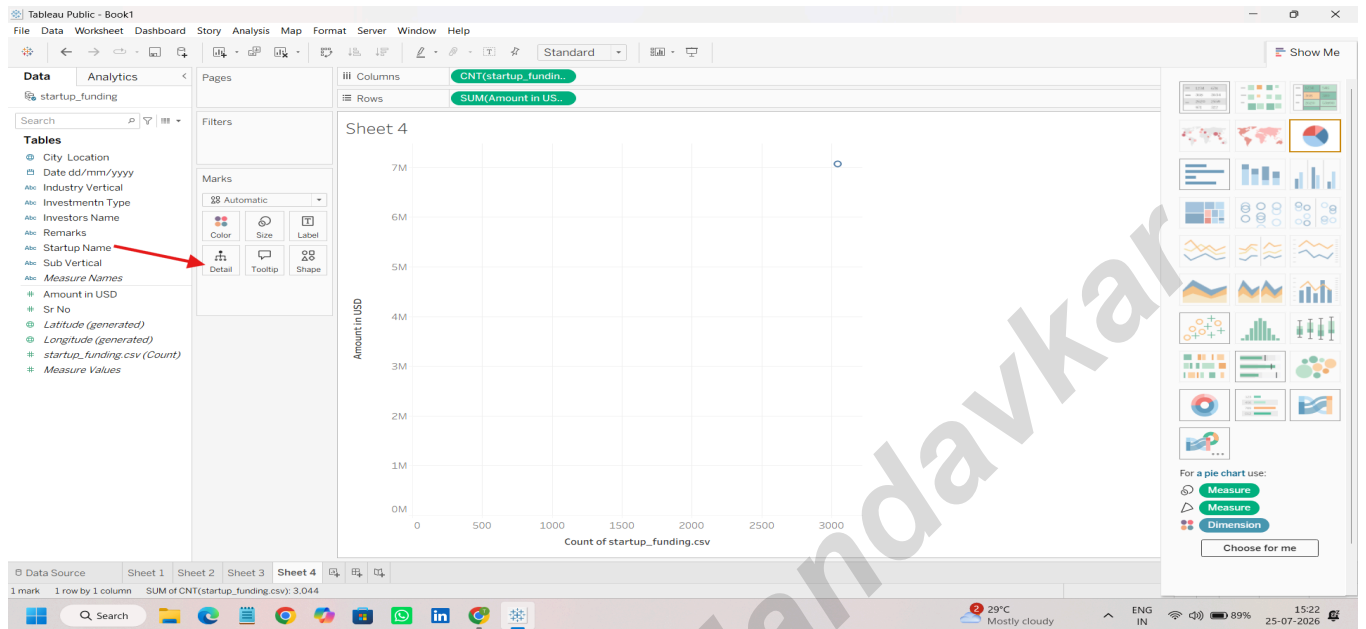
Name:- Chetan Mandavkar

Roll No. T093

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Step 6:- Drag Startup Name and drop it directly onto the Detail tile within the Marks card.



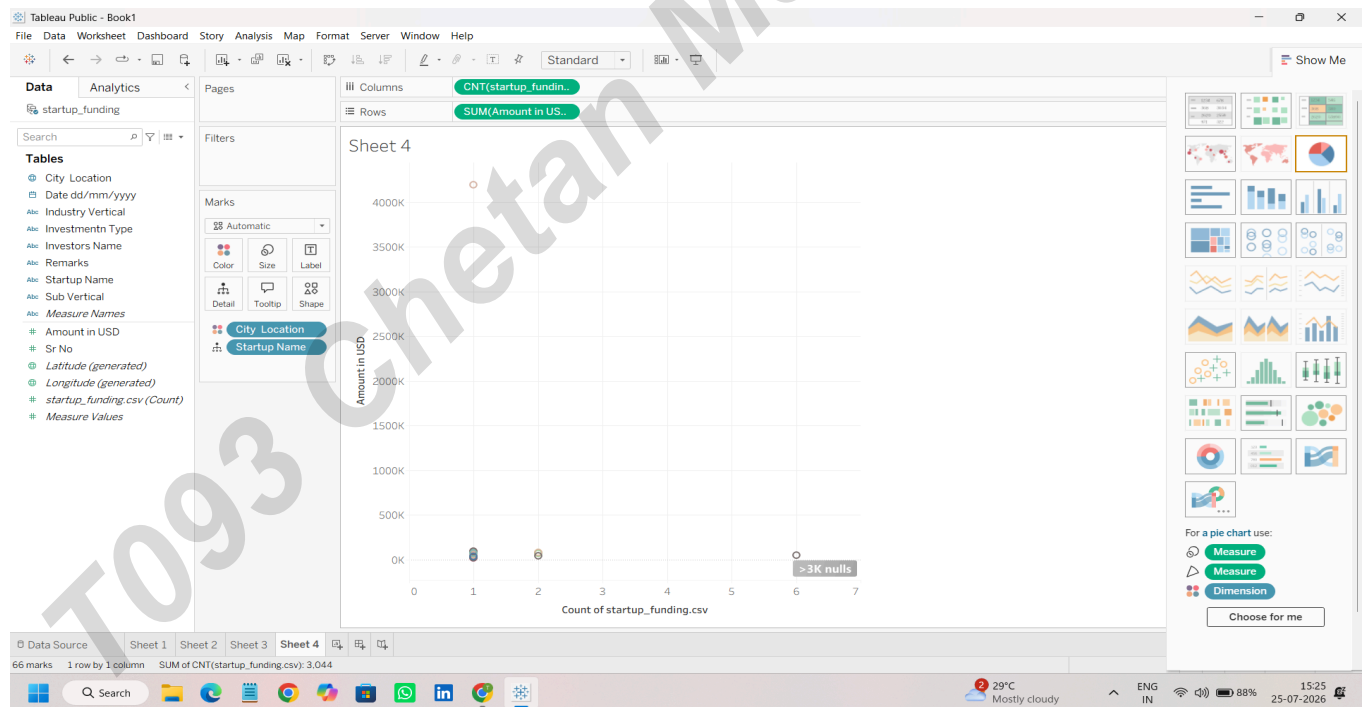
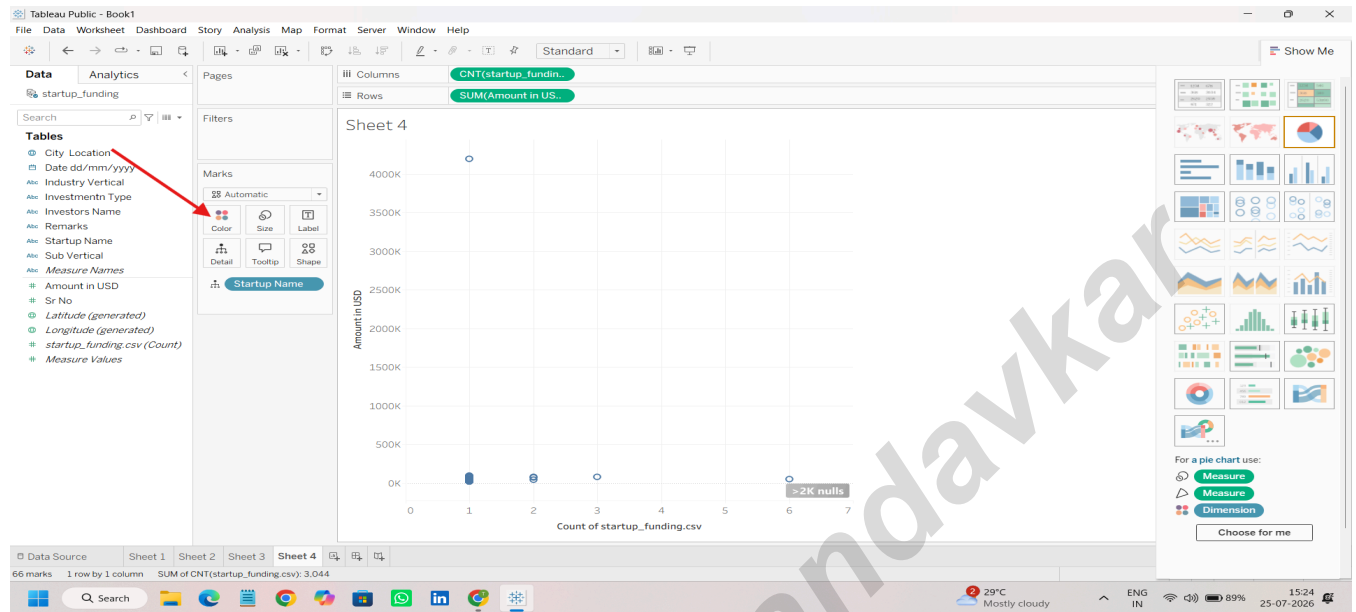
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

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Step 7:- Adding Context with Color: Take the City Location dimension from your Data Pane and drop it onto the Color tile in the Marks card.



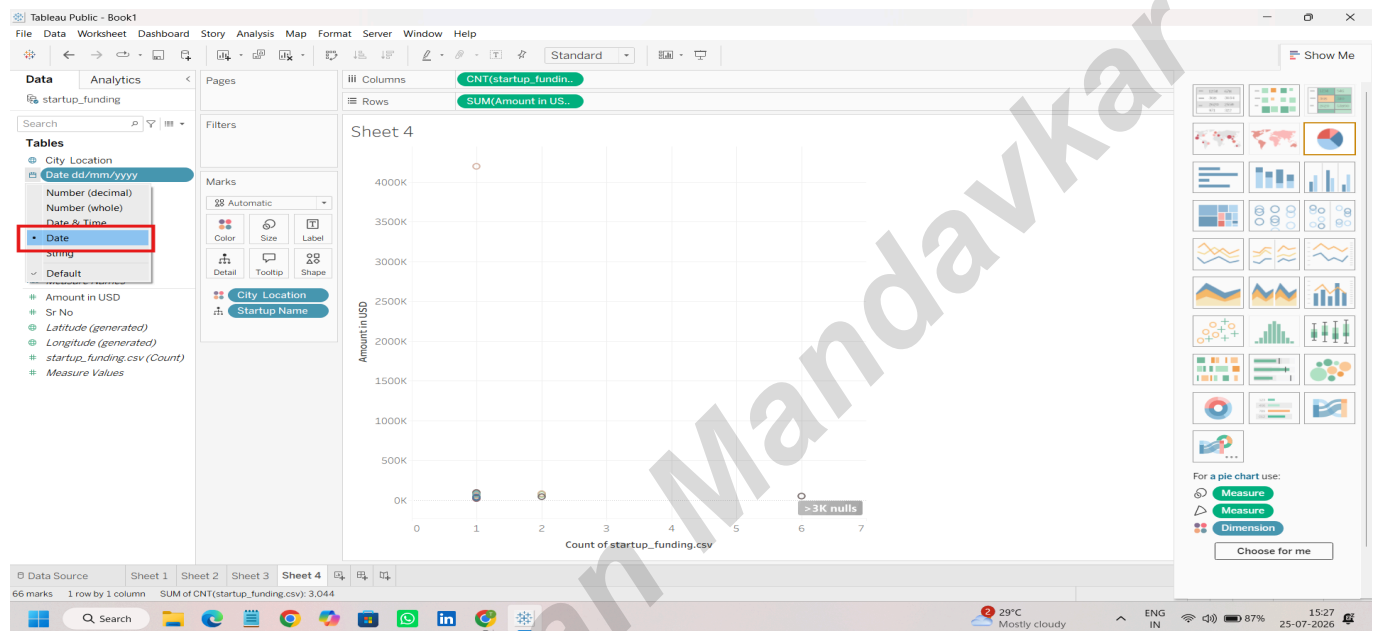
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

D] Create dual-axis and combination charts

Step 1:- Look at your left-hand Data Pane. Find the field Date dd/mm/yyyy and click the small text icon (Abc) immediately to the left of its name. Select Date from the dropdown menu to convert it into a structured time dimension.



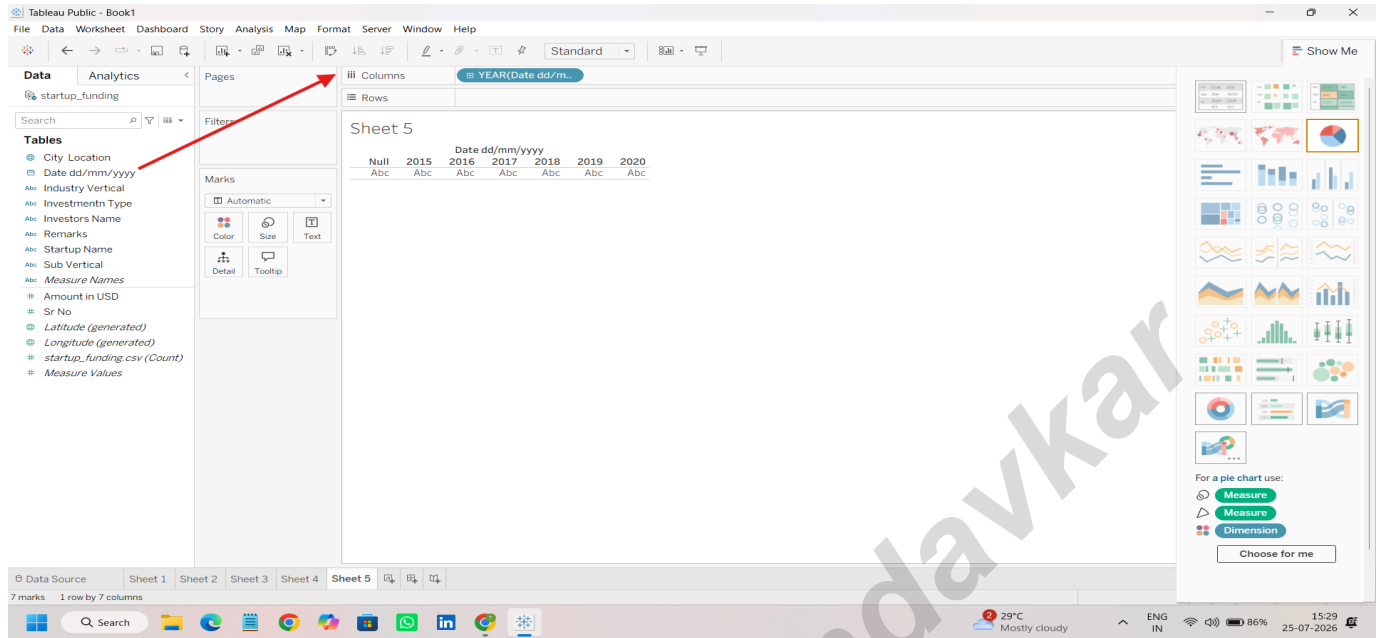
Step 2:- Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 5). Drag your newly formatted Date dd/mm/yyyy dimension and drop it onto the Columns shelf.

Name:- Chetan Mandavkar

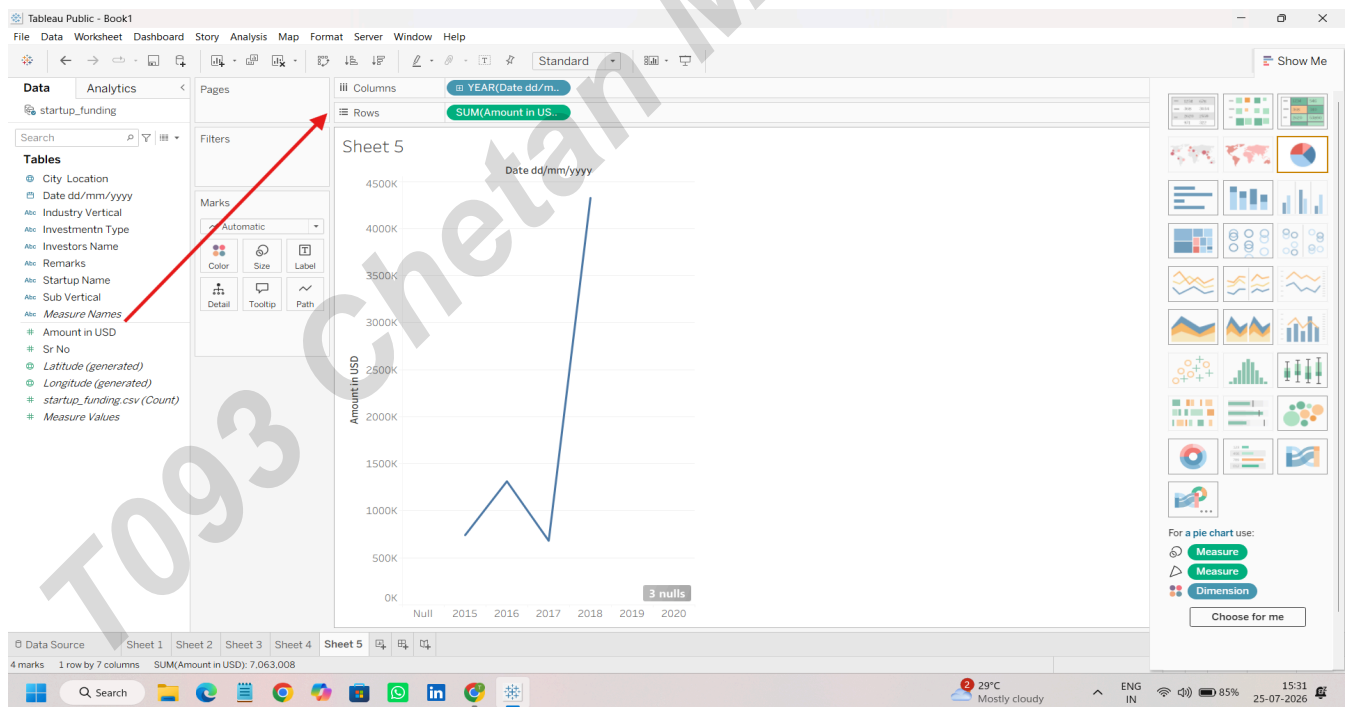
Roll No. T093

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Step 3:- Drag the Amount in USD measure from the Data Pane and drop it onto the Rows shelf.



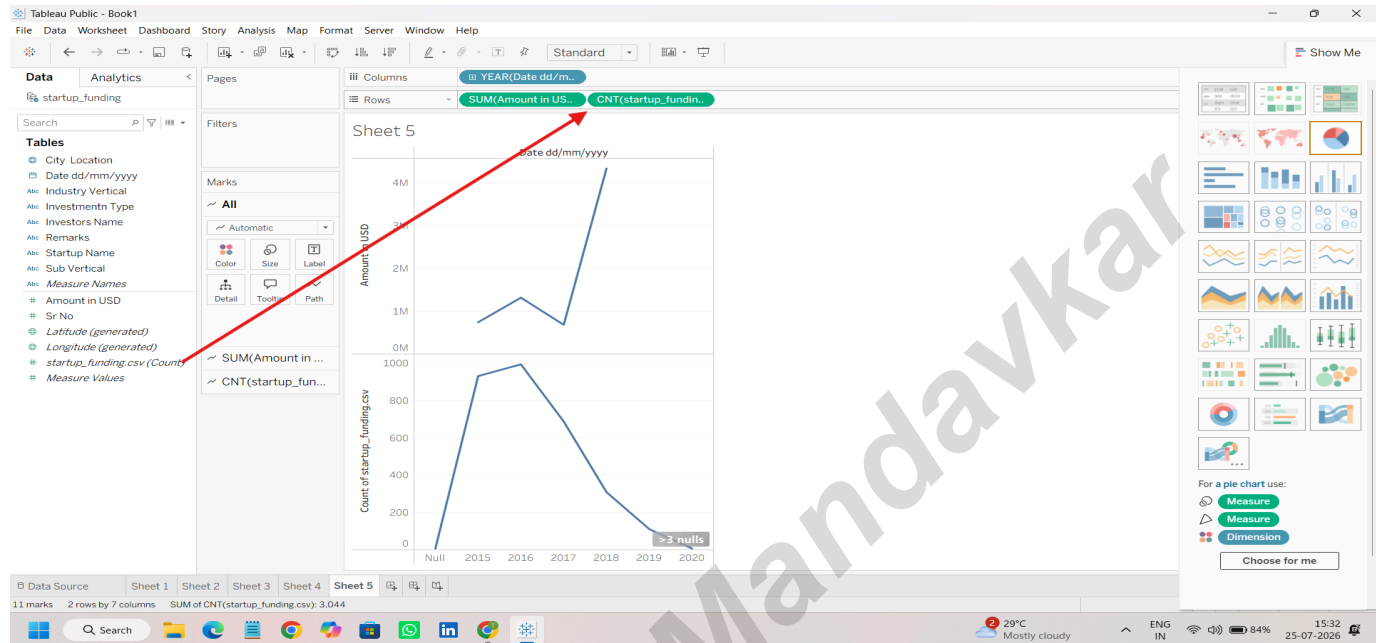
Name:- Chetan Mandavkar

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Step 4:- Now, take the automatic system measure startup_funding.csv (Count) (or Number of Records) and drag it onto the Rows shelf, placing it directly to the right of the Amount in USD pill.



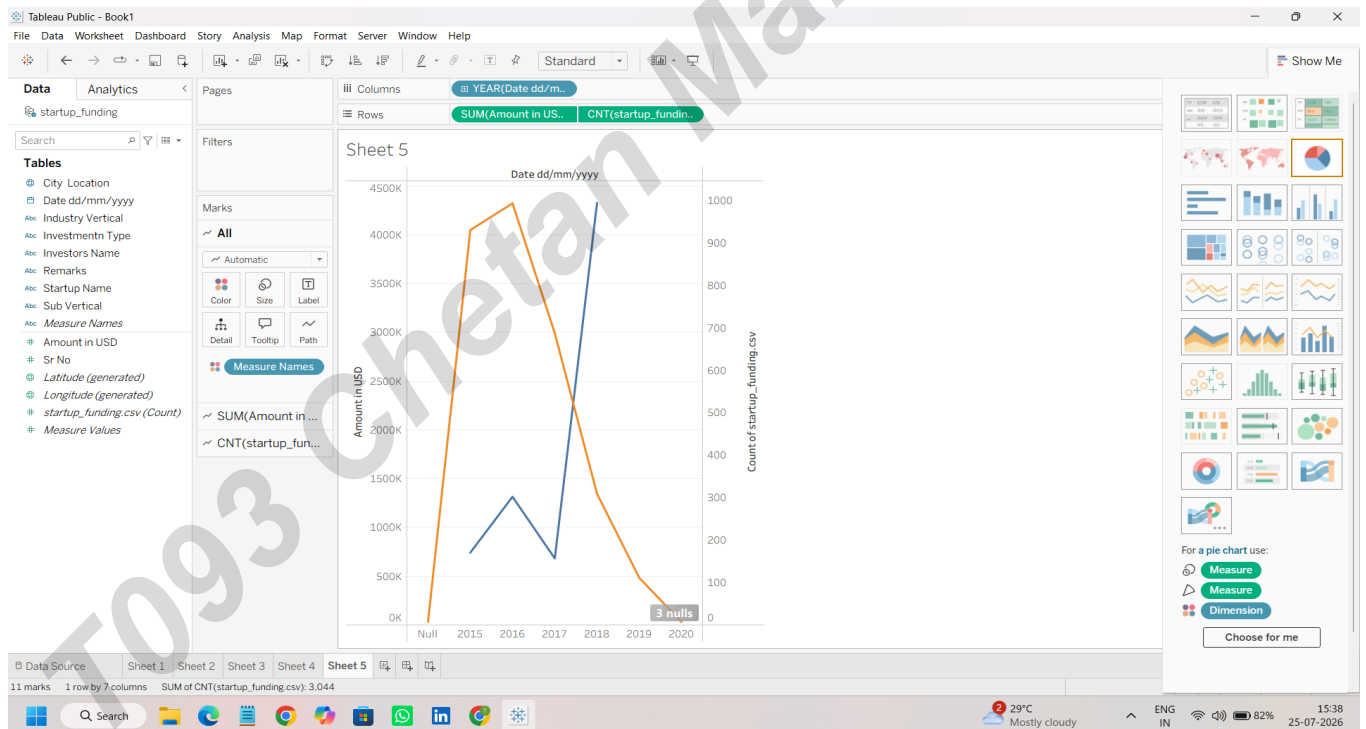
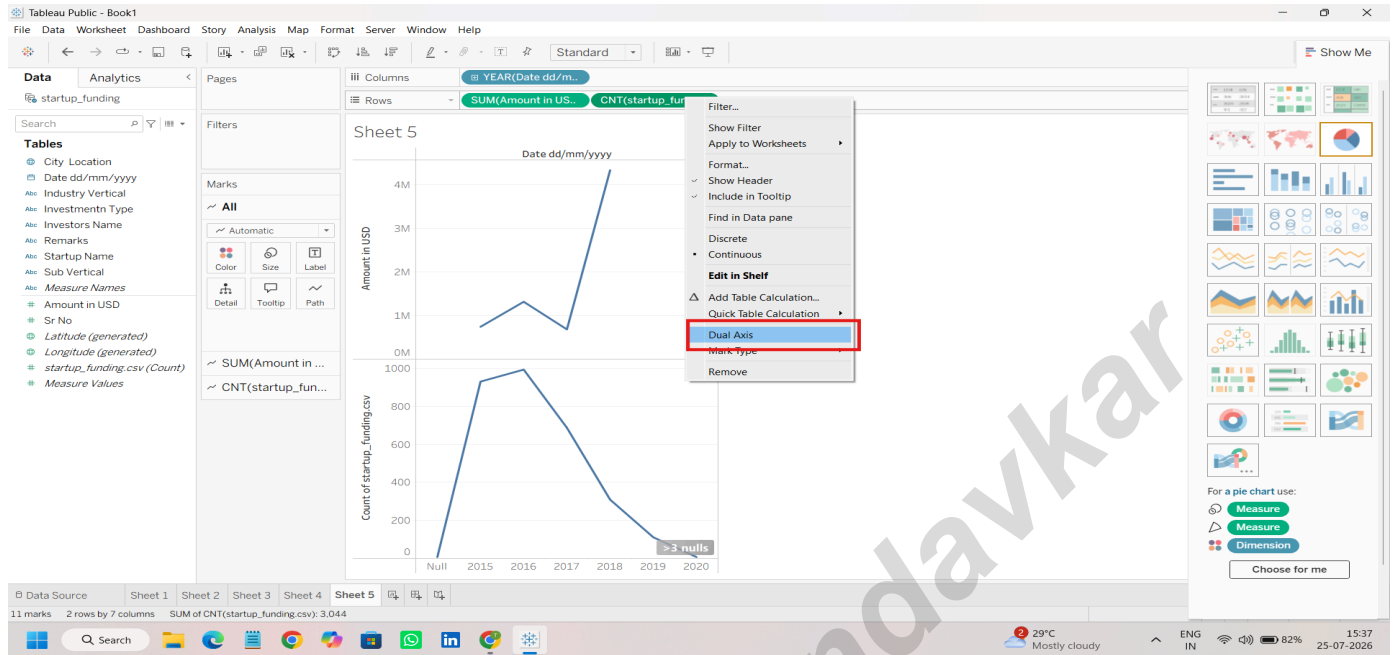
Step 5:- Right-click the second measure pill (startup_funding.csv (Count)) sitting up on the Rows shelf. Select Dual Axis from the context dropdown menu.

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Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

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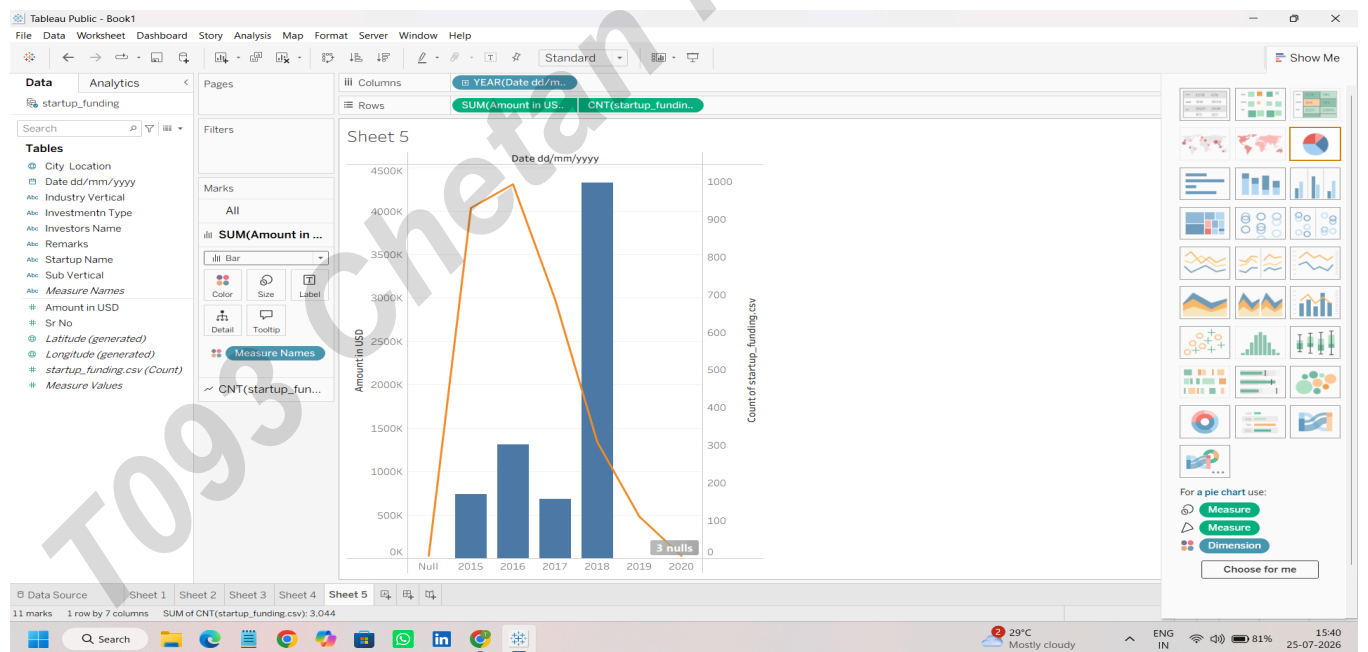
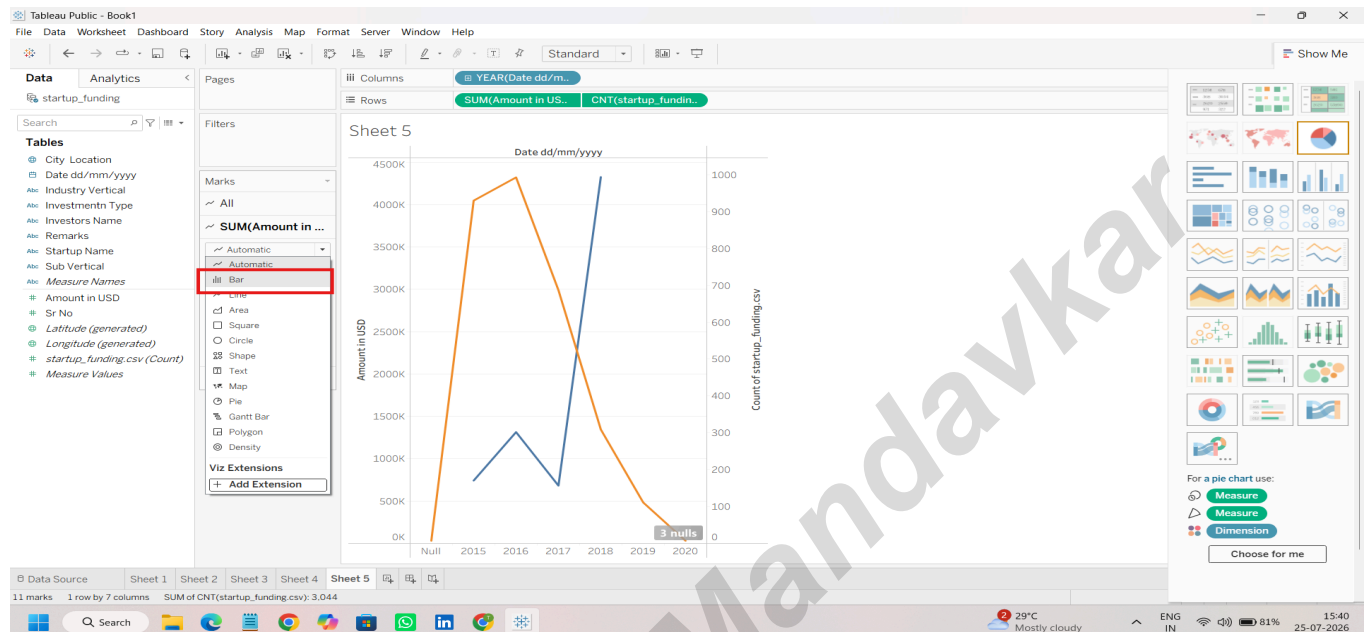
Name:- Chetan Mandavkar

Roll No. T093

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Step 6:- Click directly on the SUM(Amount in USD) tab in the Marks card. Change its visual type dropdown selection from Automatic to Bar.



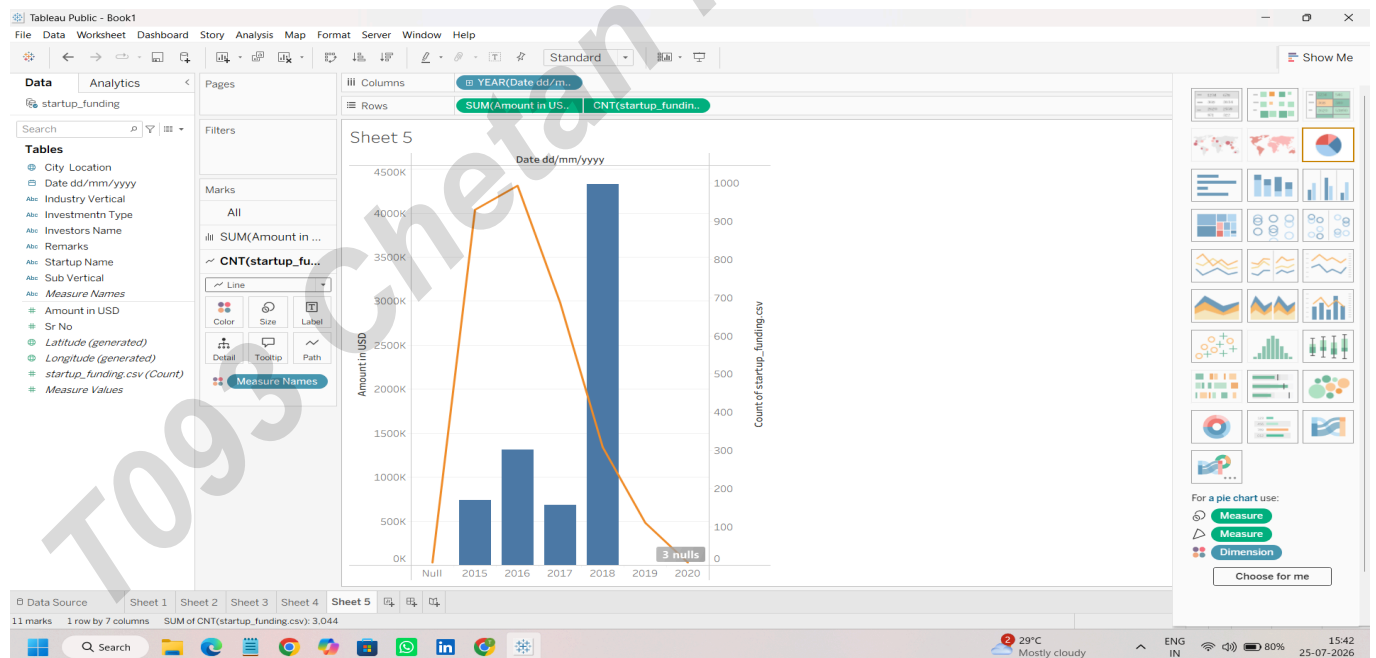
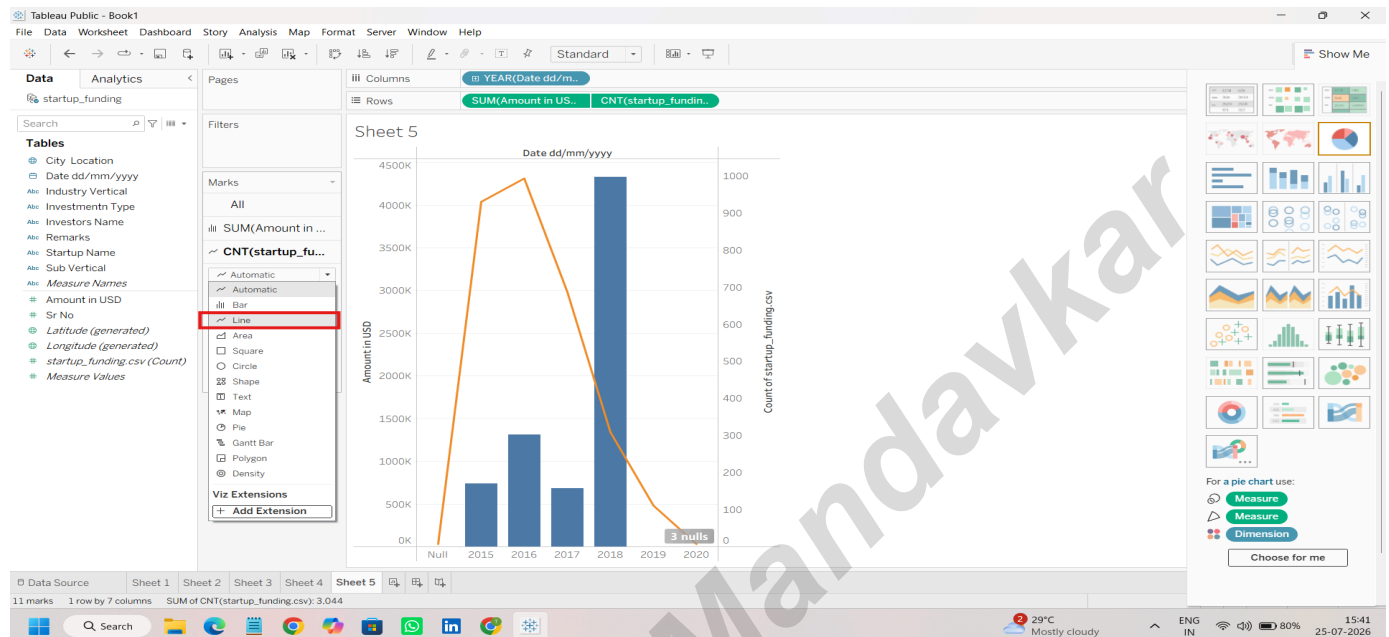
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

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Step 7:- Next, click directly on the CNT(startup_funding.csv) tab in the Marks card. Change its visual type dropdown selection from Automatic to Line.



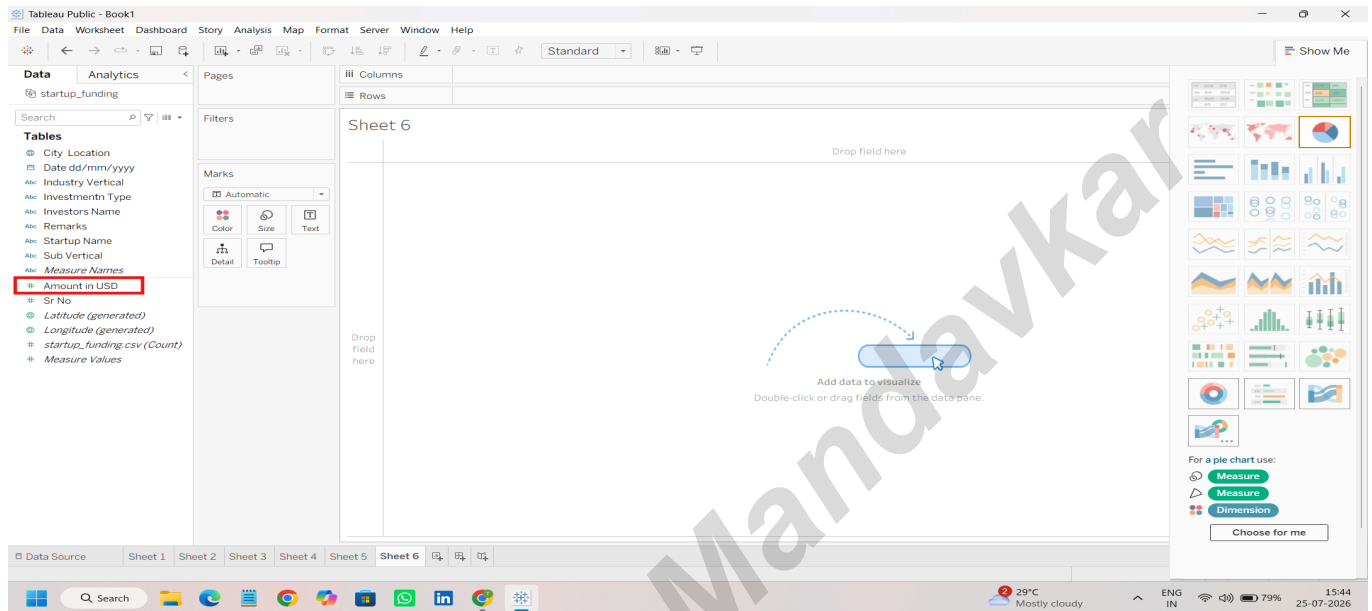
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

E: Developing Histograms for Distribution Analysis

Step 1:- Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 6). Go to the Data Pane and click once directly on the Amount in USD measure to highlight it.



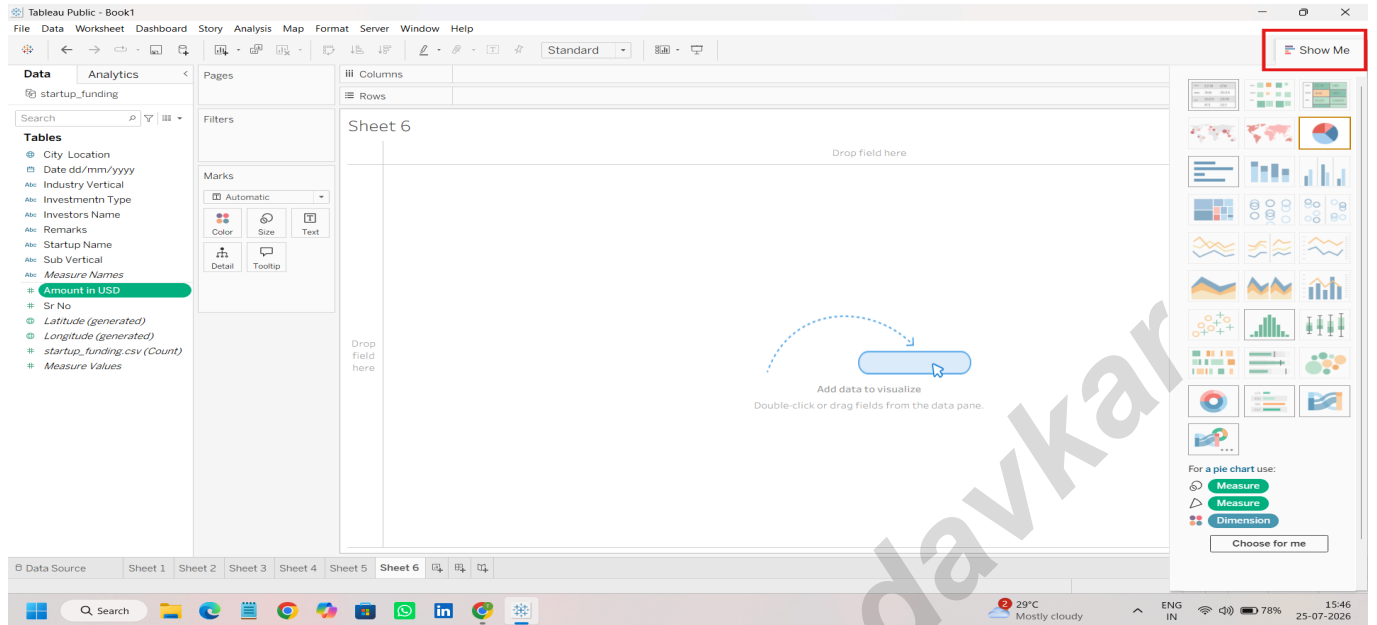
Step 2:- Move your mouse to the top-right corner of the window and click the Show Me button to expand the chart library panel.(Which is already Expanded)

Name:- Chetan Mandavkar

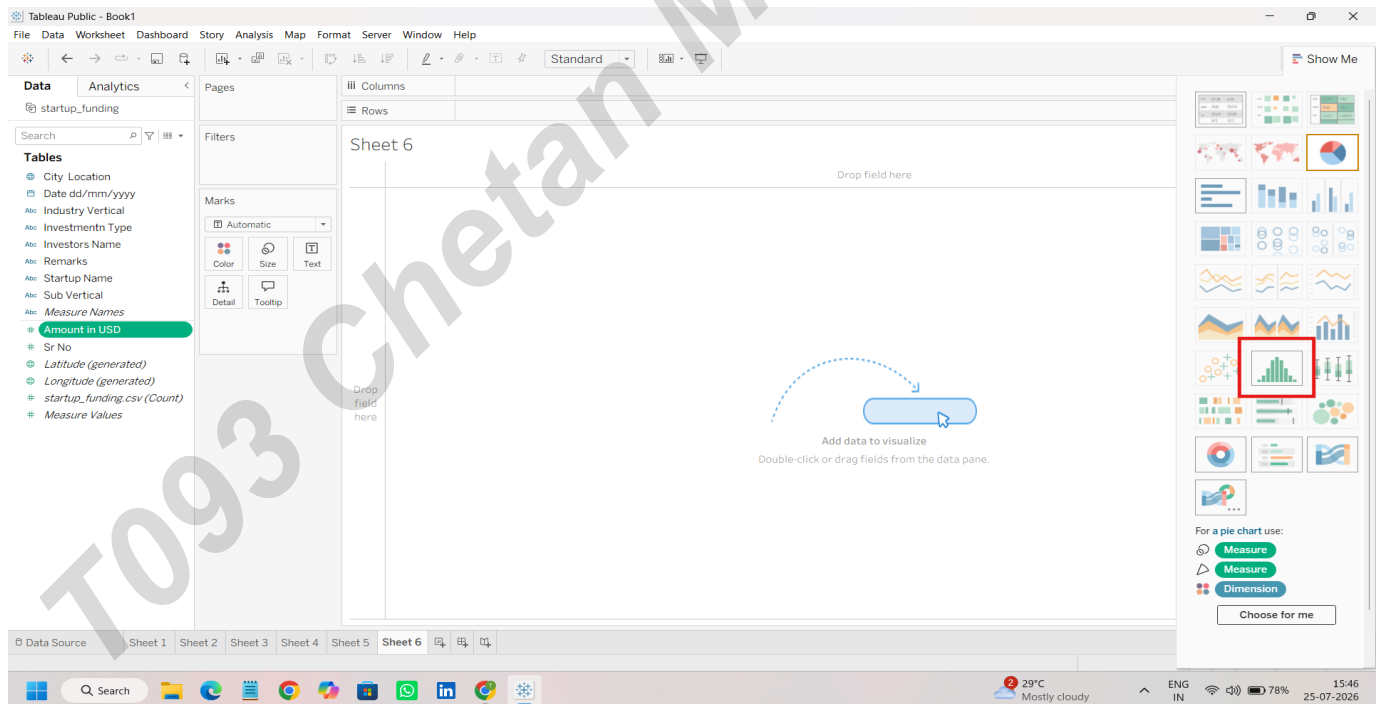
Roll No. T093

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Step 3:- From the options, locate and click the Histogram icon (found in the middle row, displaying vertical green bars).

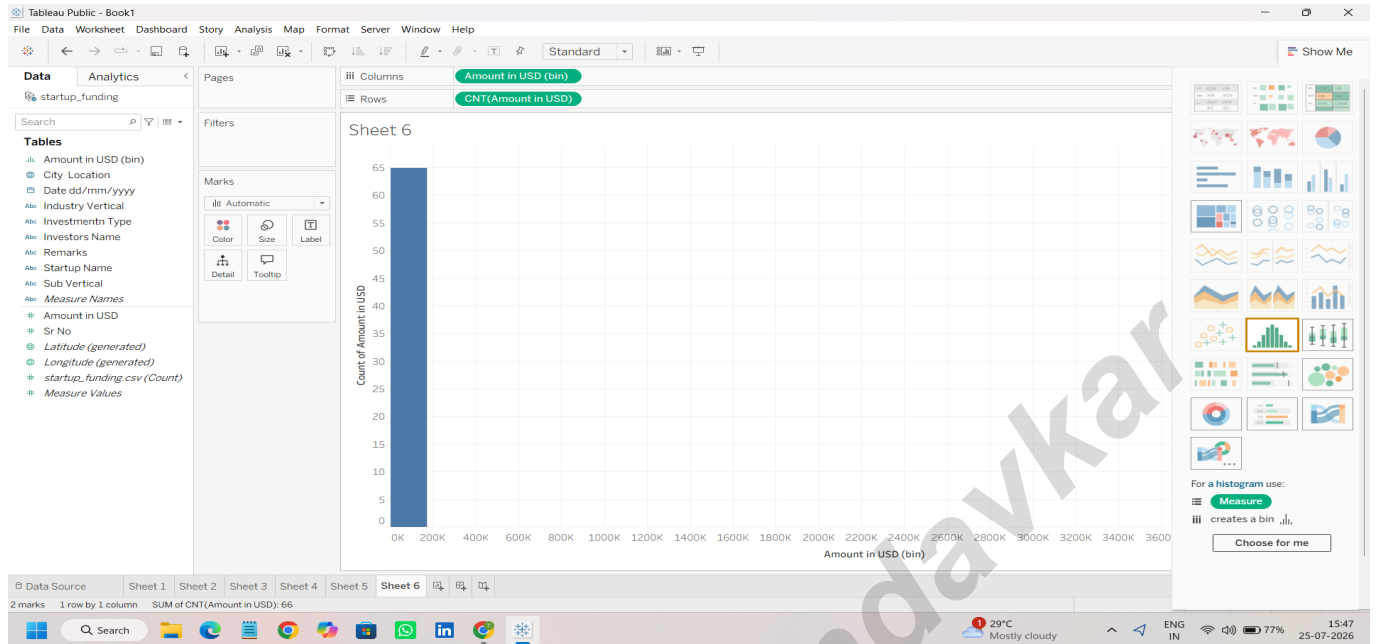


Name:- Chetan Mandavkar

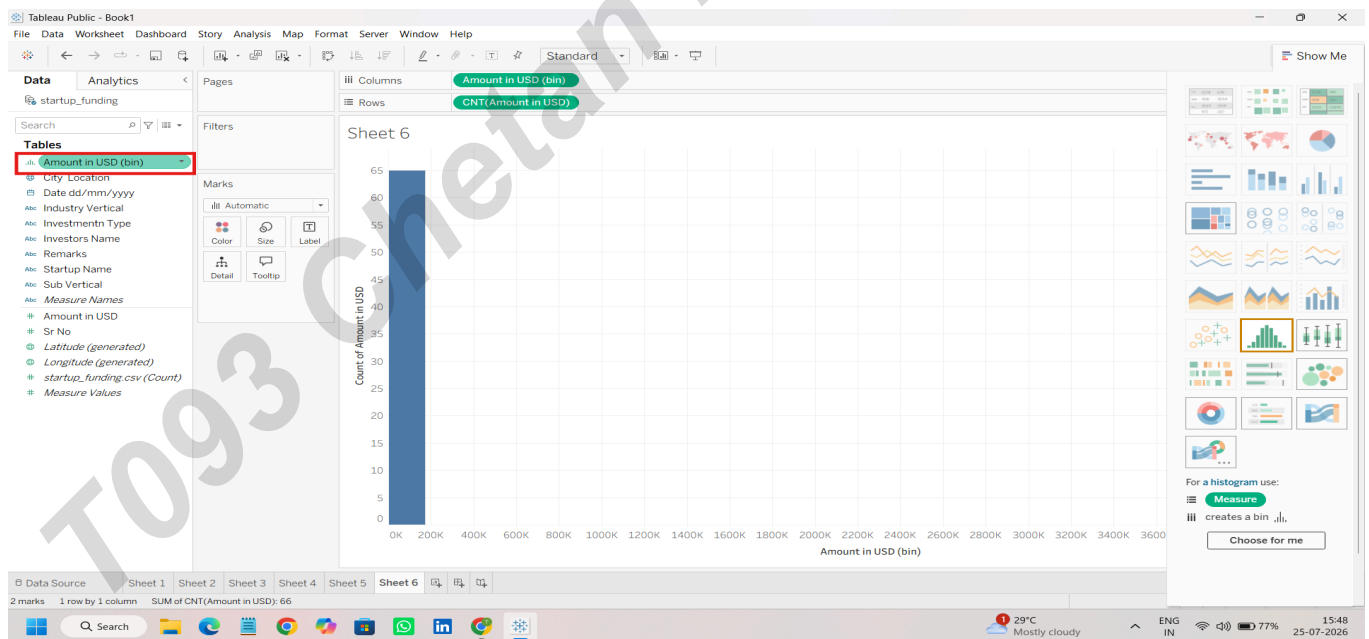
Roll No. T093

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Step 4:- Look back at your Data Pane. Notice that Tableau has automatically generated a brand-new field under your dimensions list called Amount in USD (bin) and placed it onto your Columns shelf.



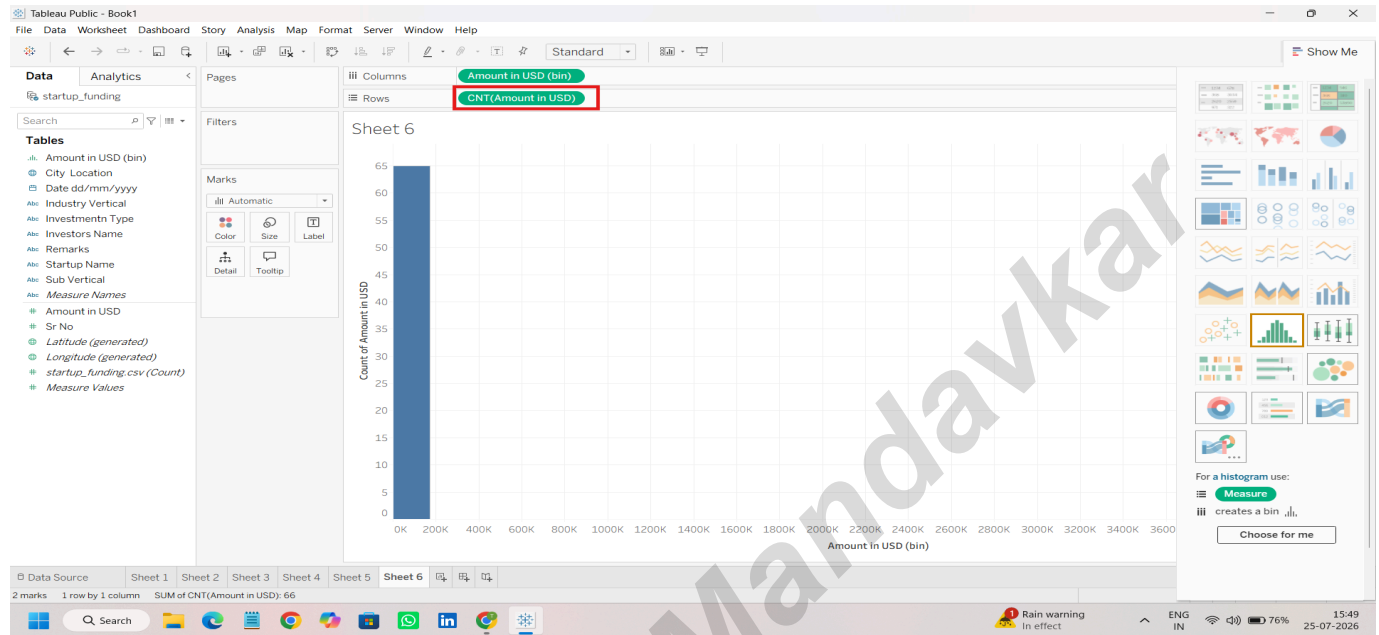
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

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Step 5:- Look at the Rows shelf. Tableau has automatically placed the aggregated measure CNT(Amount in USD) there to track record counts.



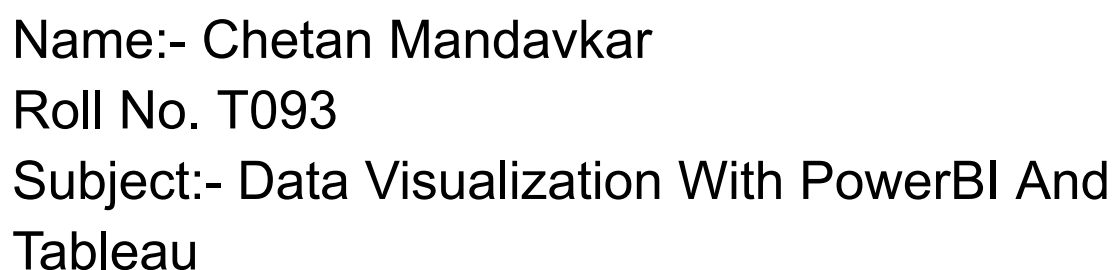
F: Building Box-and-Whisker Plots

Step 1:- Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 7) Drag the City Location dimension from the Data Pane and drop it onto the Columns shelf.

Name:- Chetan Mandavkar

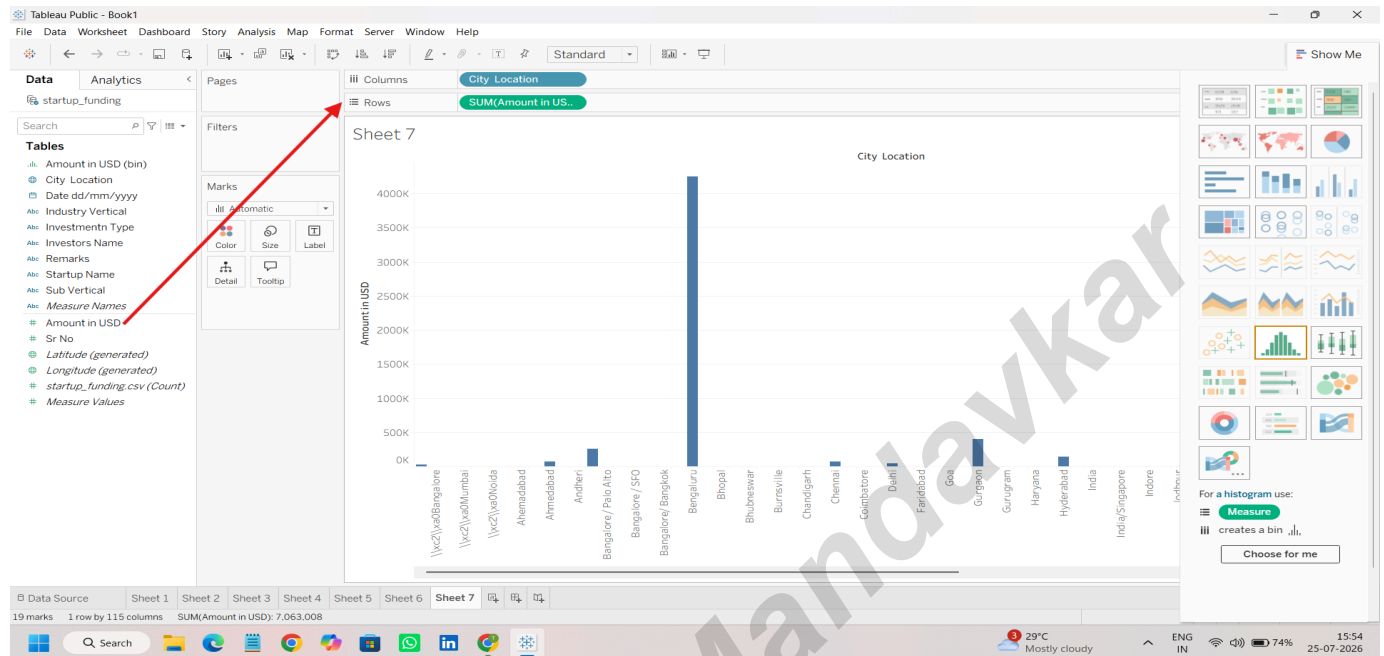
Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

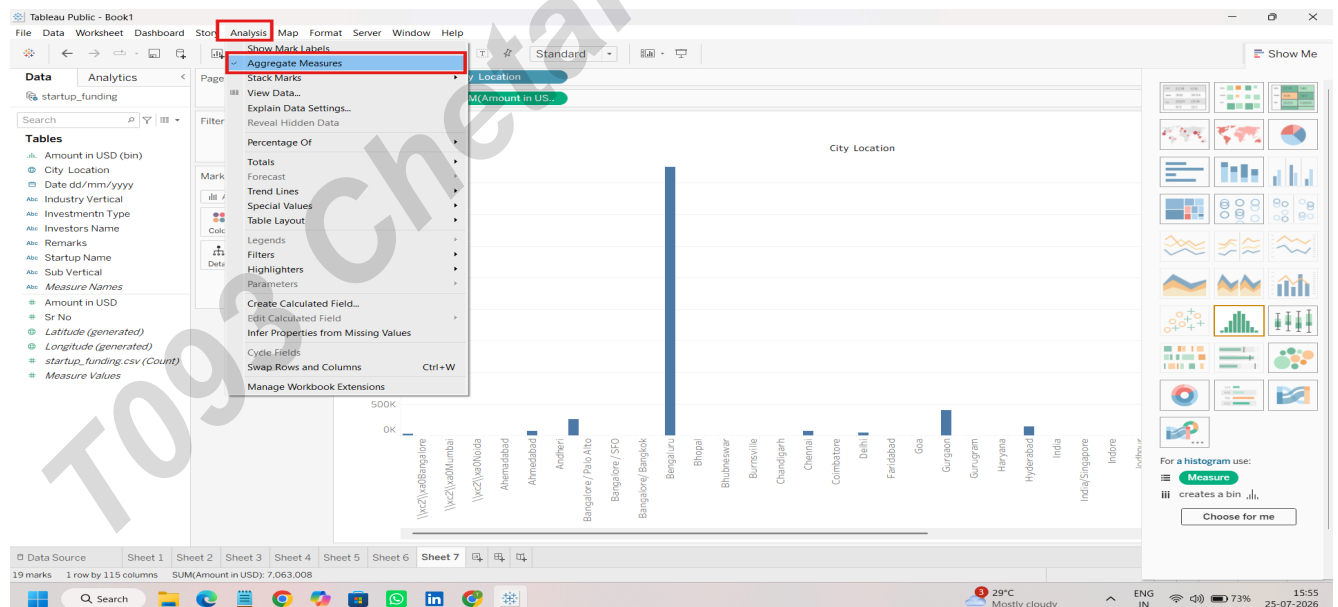


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Step 2:- Drag the Amount in USD measure and drop it onto the Rows shelf.



Step 3:- Disaggregate the Data Points: Go to the top menu bar click Analysis, and uncheck Aggregate Measures.

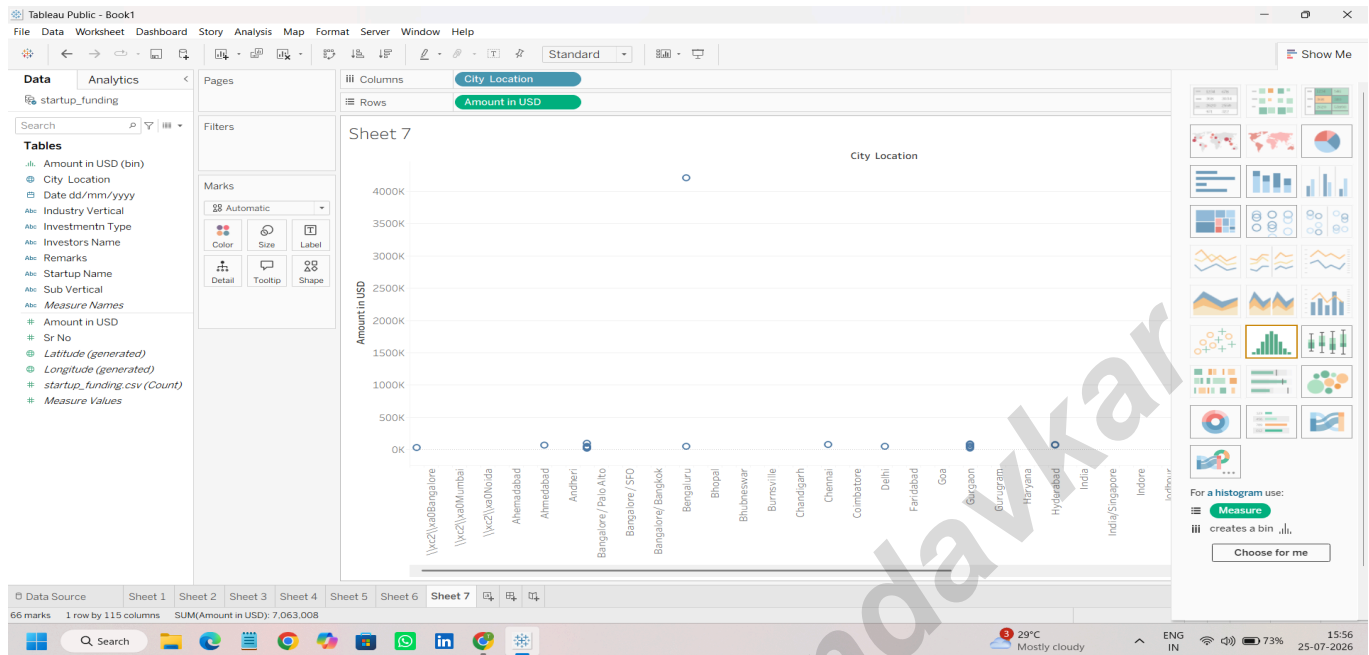


Name:- Chetan Mandavkar

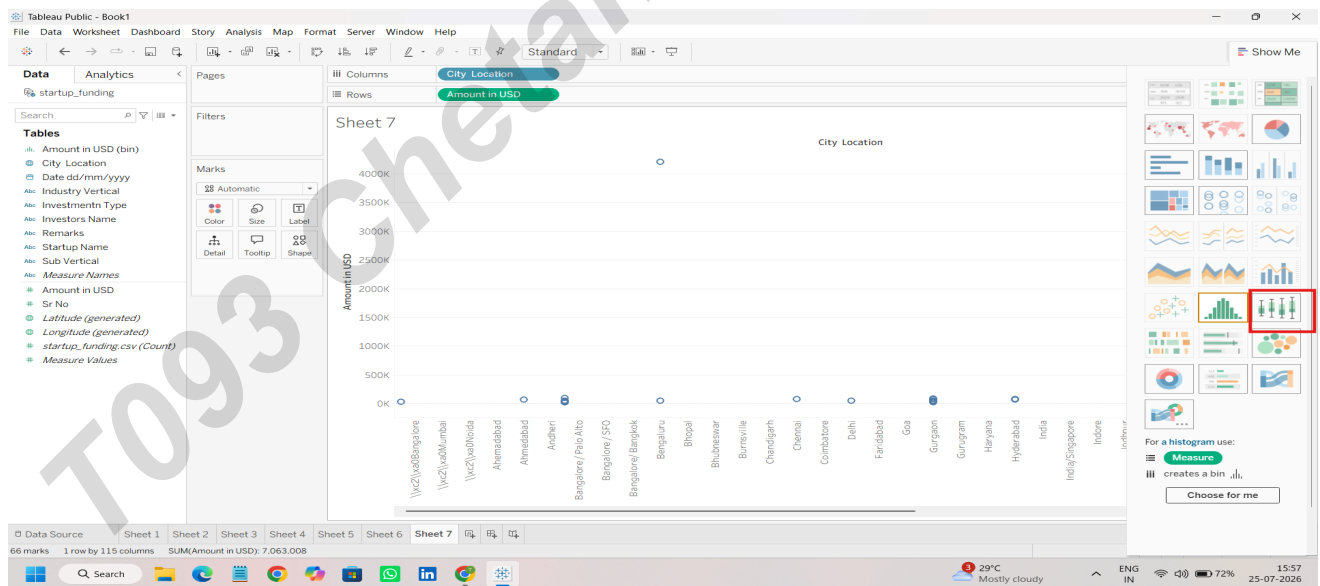
Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

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Step 4:- Add the Box Plot: Look at the top-right corner of the window, click the Show Me button, and select the Box-and-Whisker Plots icon (found in the middle row, showing a box shape with extended antenna lines).

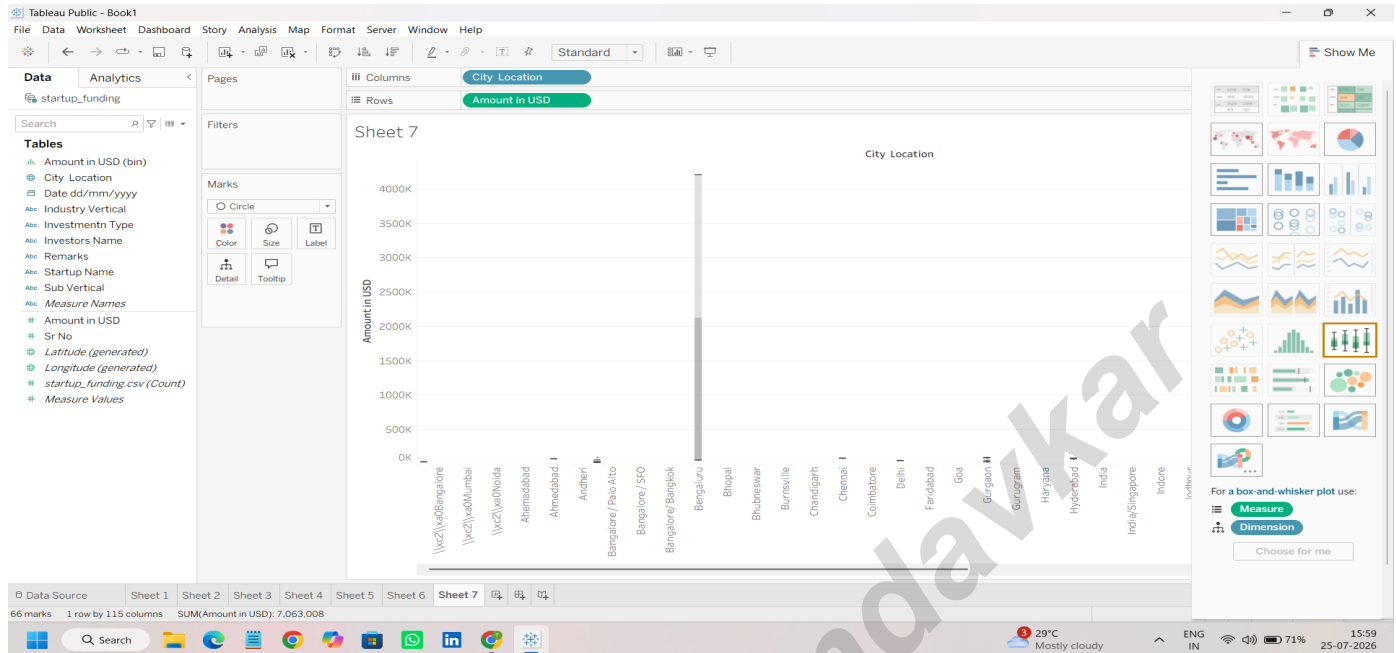


Name:- Chetan Mandavkar

Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

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The screenshot displays the Tableau Public interface. The main view is a box-and-whisker plot titled 'Sheet 7' showing the distribution of 'Amount in USD' (Y-axis) across different 'City Location' categories (X-axis). The X-axis categories include Bangalore/SFO, Bangalore/Bangkok, Bhubaneswar, Indore, Jodhpur, Karnataka, Kerala, Kolkata, Kormangala, London, Marlo Park, Mumbai, Mumbai/NY, and Mumbai/Bangalore. A large, semi-transparent watermark 'Indavkar' is overlaid diagonally across the plot area.

A 'Filter [City Location]' dialog box is open in the center. It has tabs for 'General', 'Wildcard', 'Condition', and 'Top'. The 'General' tab is active, showing options to 'Select from list', 'Custom value list', or 'Use all'. Below this is a search bar 'Enter search text' and a list of cities with checkboxes:

- ☐ Belgium
- ☐ Bengaluru
- ☐ Bengaluru and Gurugram
- ☐ Bhopal
- ☐ Bhubaneswar
- ☒ Bhubaneswar
- ☒ Boston
- ☒ Burnsville
- ☒ California
- ☒ Chandigarh
- ☒ Chembur

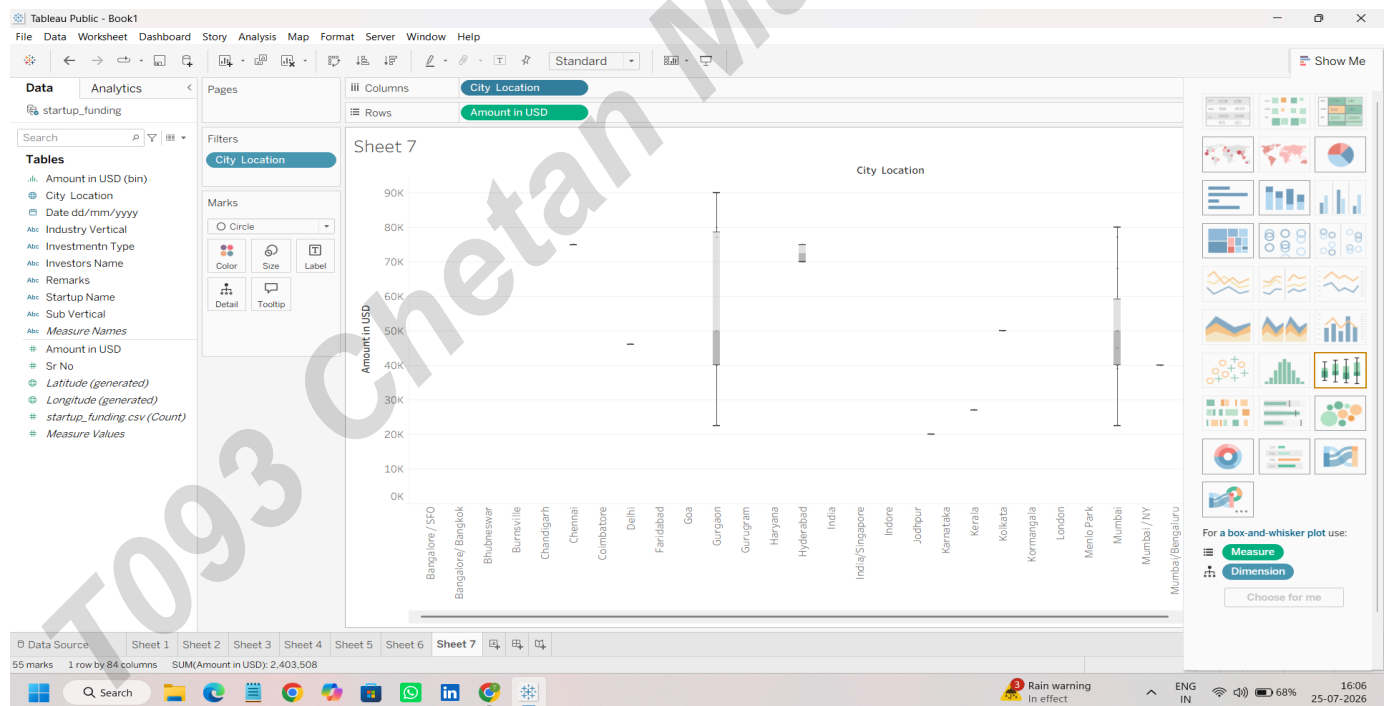
 At the bottom of the dialog, there are buttons for 'All', 'None', and 'Exclude', along with a 'Summary' section showing:

- Field: [City Location]
- Selection: Selected 84 of 115 values
- Wildcard: All
- Condition: None
- Limit: None

 Buttons for 'Reset', 'OK', 'Cancel', and 'Apply' are at the bottom.

On the right side of the interface, there is a 'Show Me' panel with various chart templates. Below it, a section titled 'For a box-and-whisker plot use:' shows 'Measure' and 'Dimension' buttons, with a 'Choose for me' button below them.

The bottom status bar shows '55 marks', '1 row by 84 columns', and 'SUM(Amount in USD): 2,403,508'.

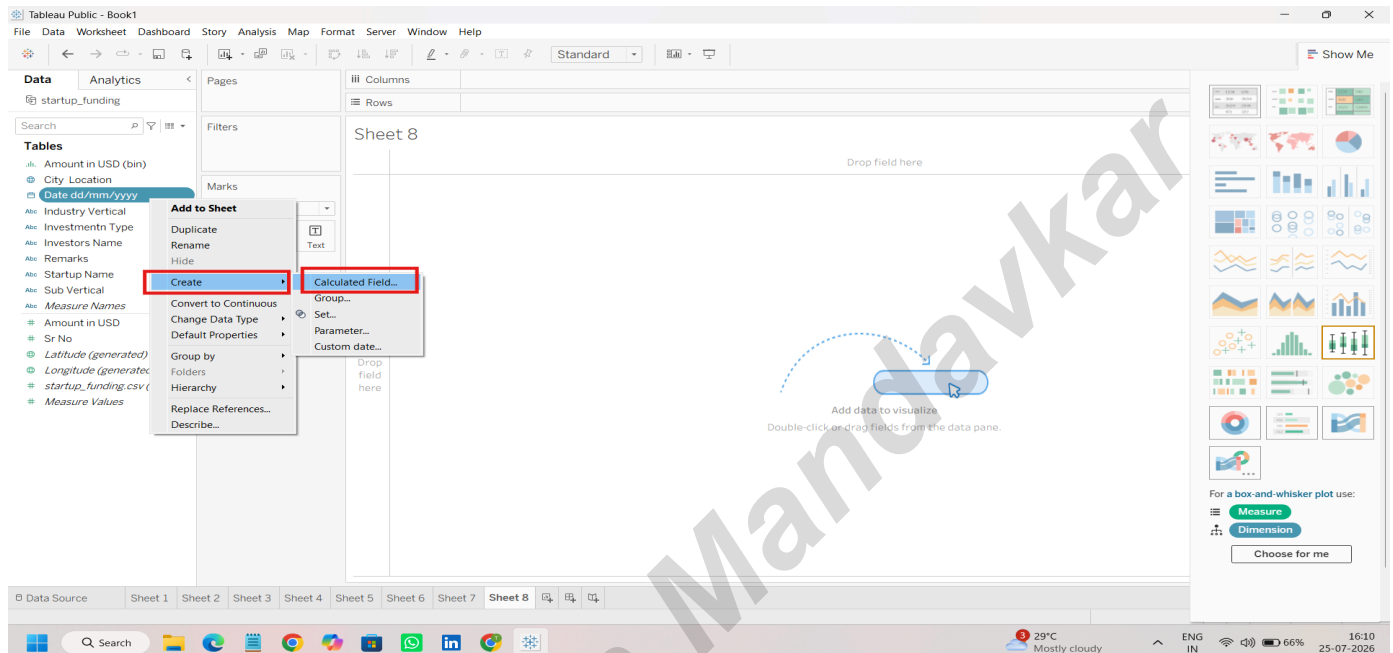


Roll No. T093

Subject:- Data Visualization With PowerBI And Tableau

G: Using Gantt Charts and Advanced Date Visualizations

Step 1:- Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 8). Look at your left Data Pane. Right-click on your Date dd/mm/yyyy field and select Create > Calculated Field.



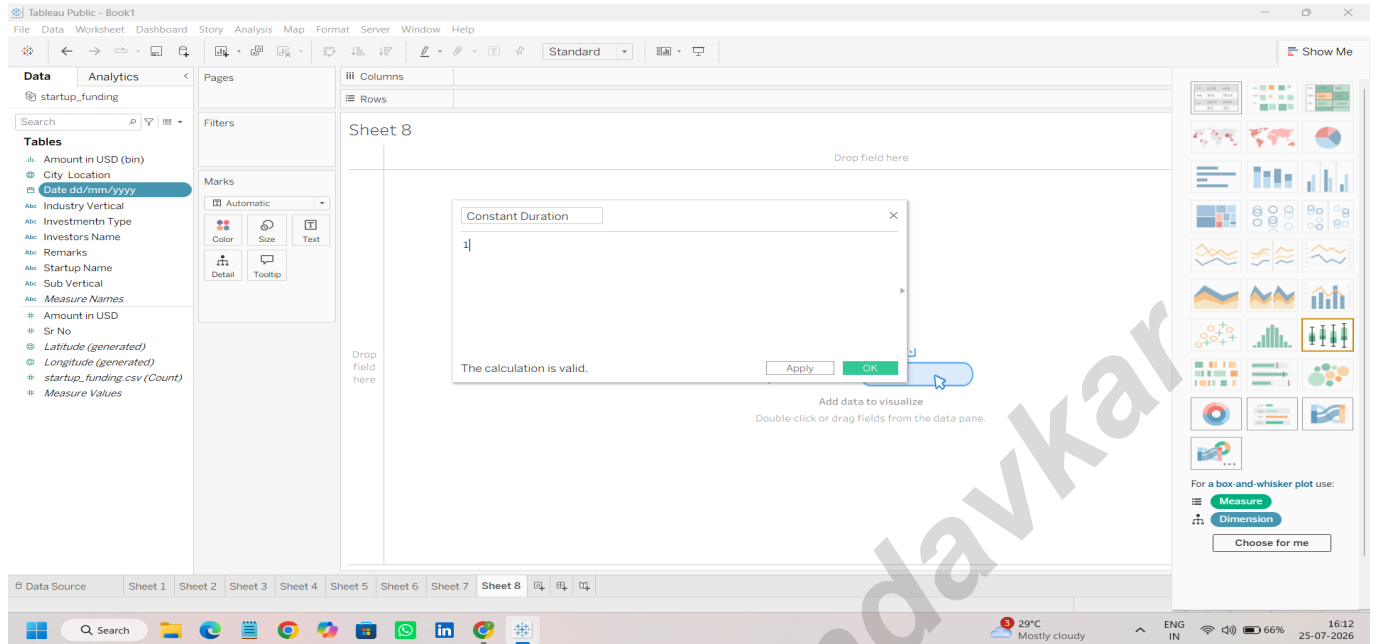
Step 2:- Name this new calculation Constant Duration. In the calculation box, type simply: 1 (This acts as a uniform placeholder length for our timeline blocks). Click OK.

Name:- Chetan Mandavkar

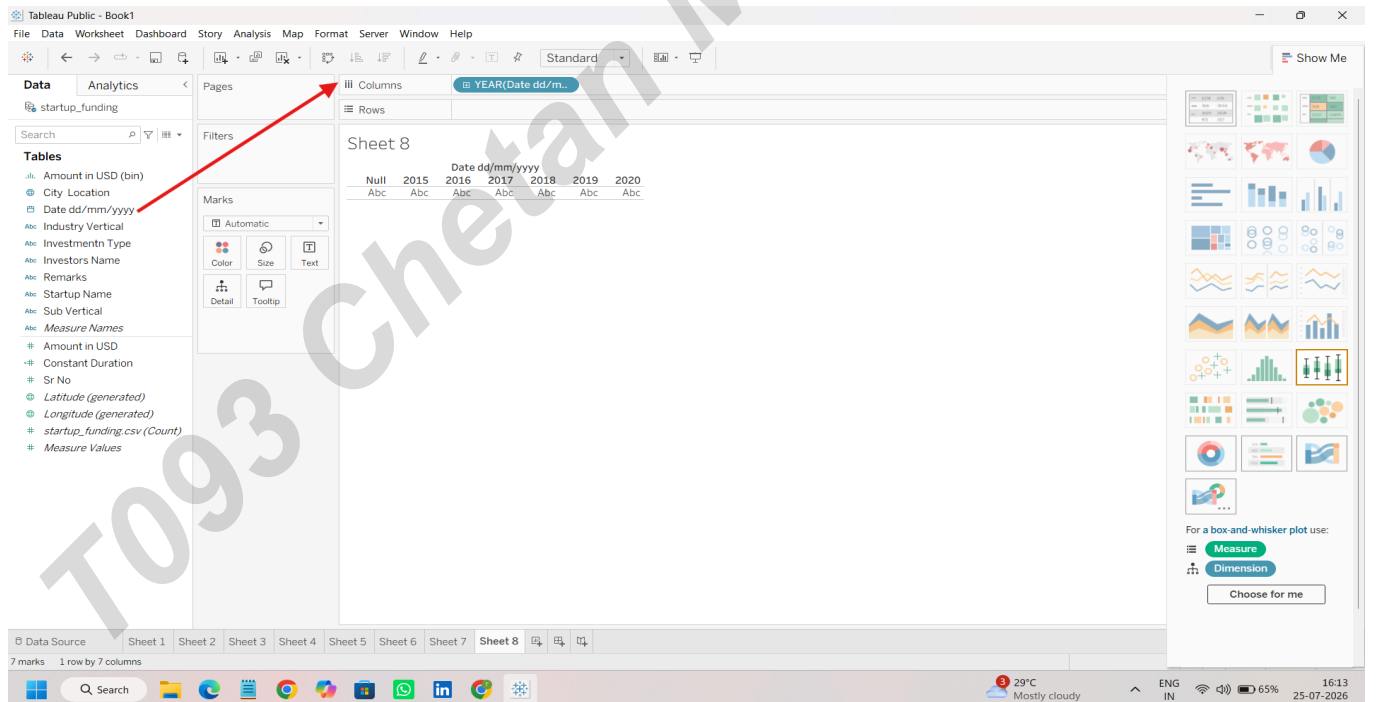
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Step 3:- Drag your continuous Date dd/mm/yyyy field and drop it onto the Columns shelf.



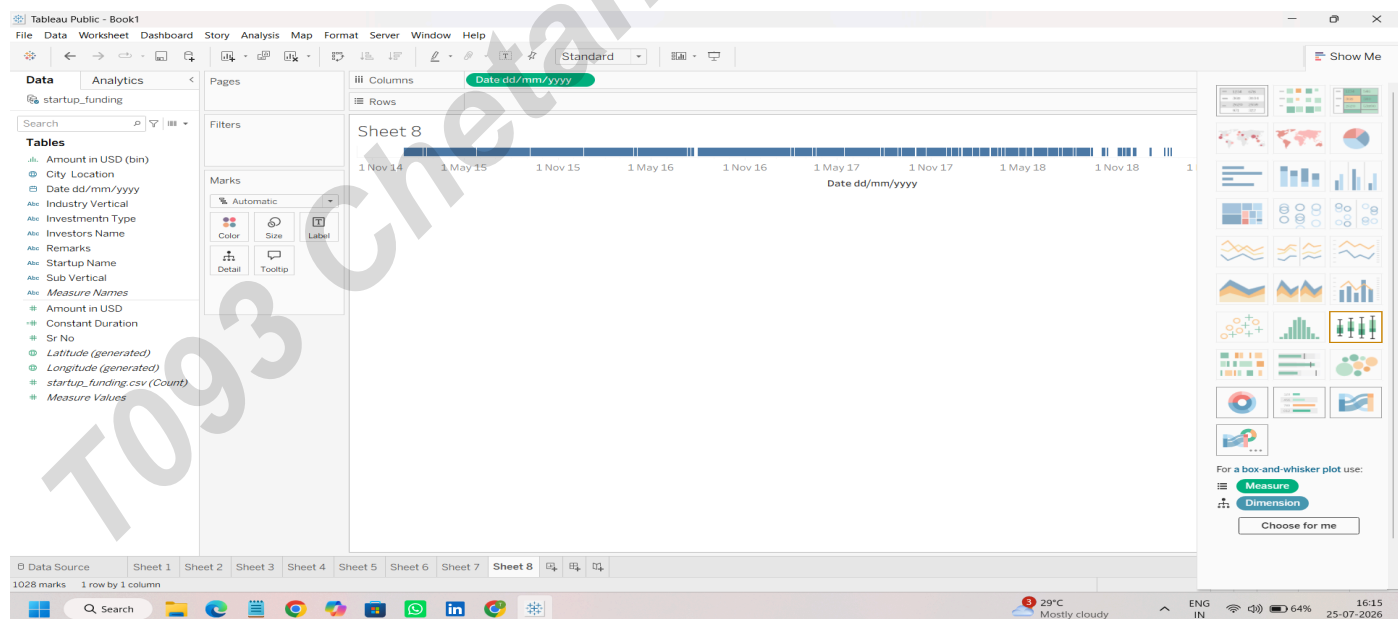
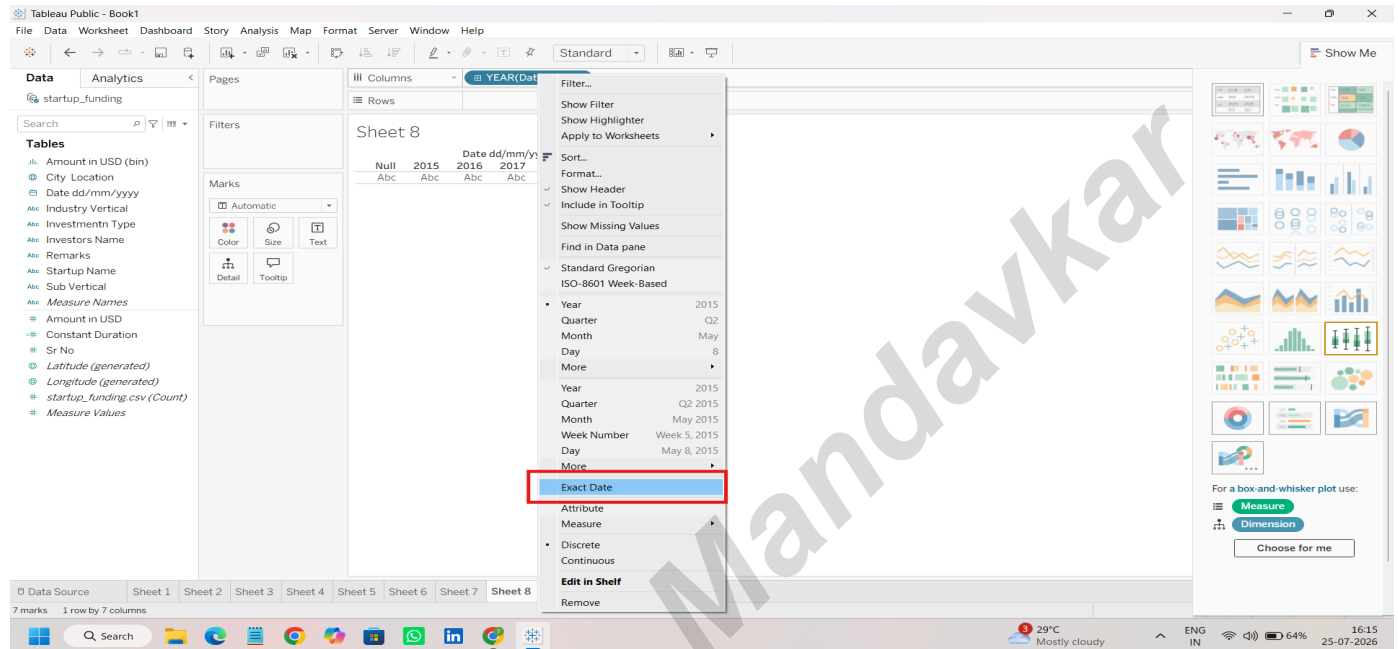
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Step 4:- Adjustment: If Tableau groups it automatically as a discrete year pill (blue YEAR(Date) pill), right-click that pill on the Columns shelf and select the continuous Exact Date option from the middle section of the dropdown menu. The pill color will change to green.



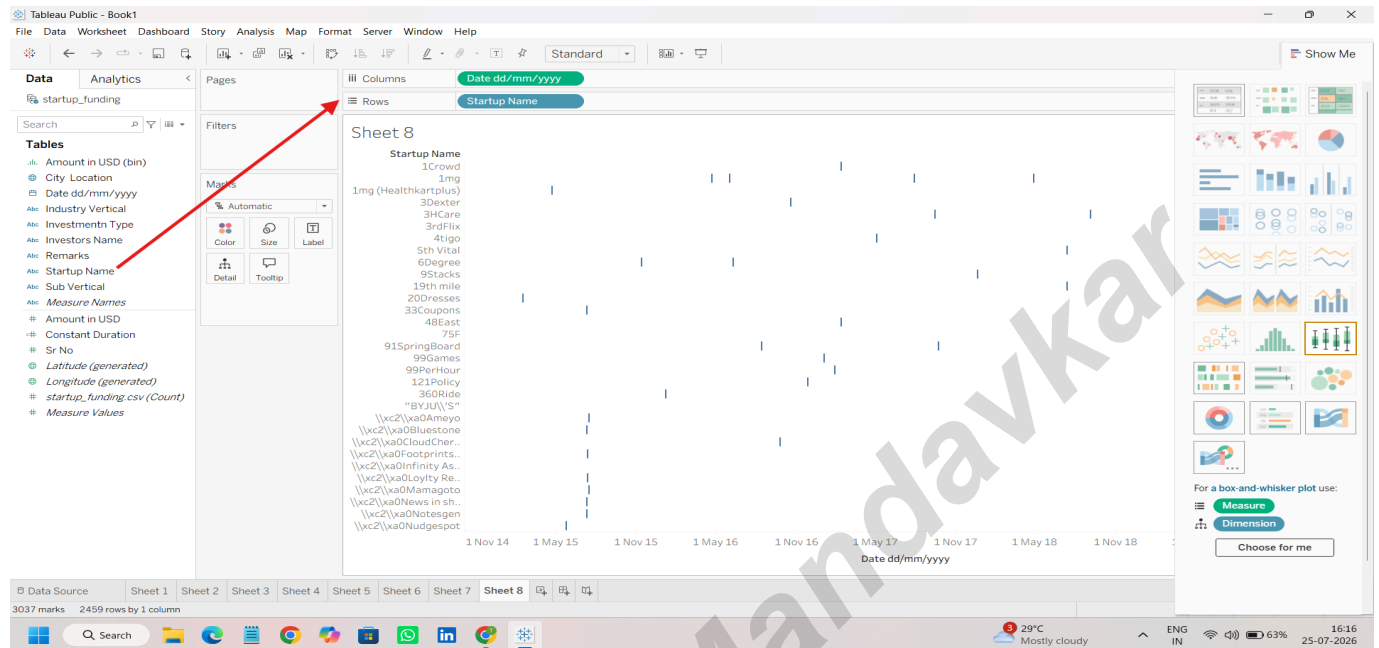
Name:- Chetan Mandavkar

Roll No. T093

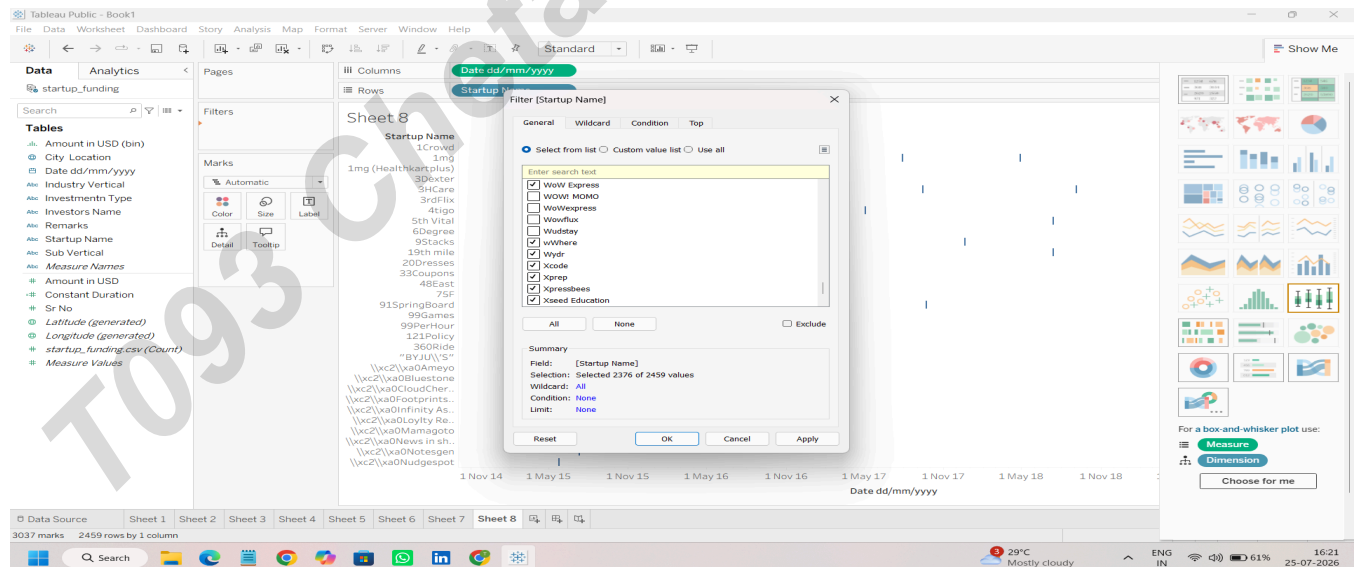
Subject:- Data Visualization With PowerBI And Tableau

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Step 5:- Drag your Startup Name dimension from the Data Pane and drop it onto the Rows shelf.



Step 6:- Apply the Filter: Because there are thousands of startups, let's look at a focused subset. Drag Startup Name into your Filters shelf, select a few notable Indian tech giants

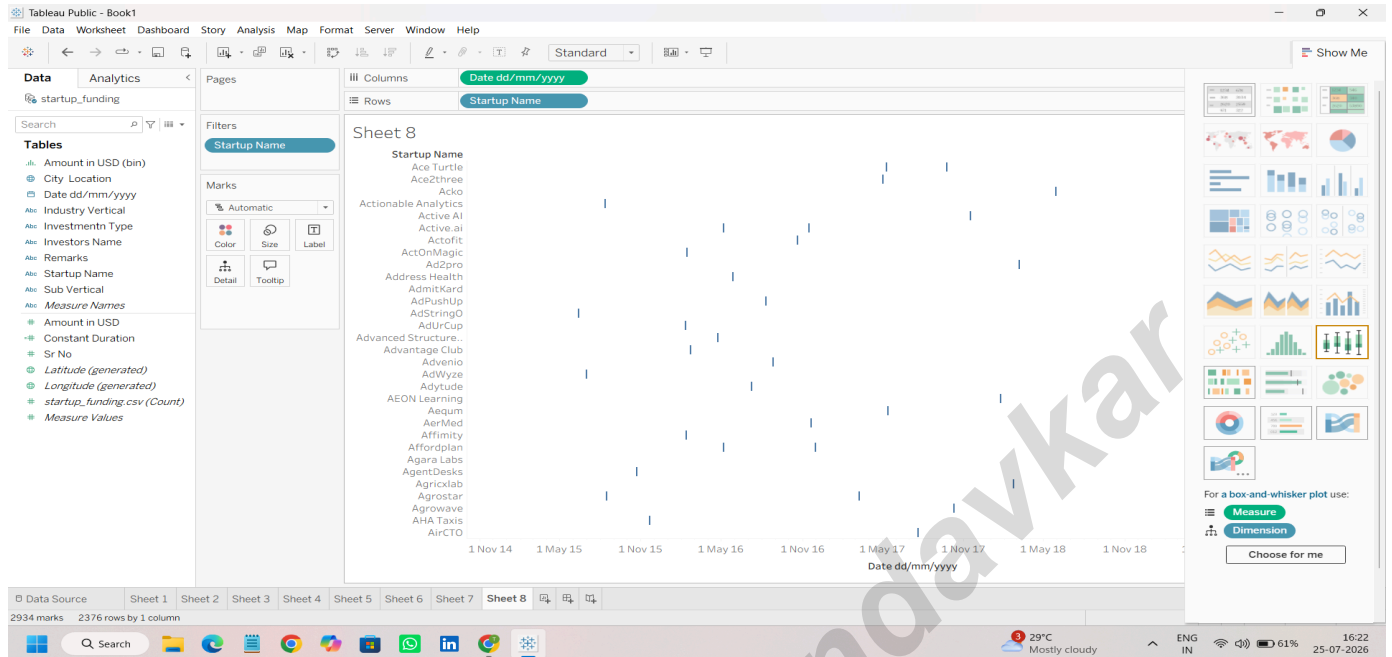


Name:- Chetan Mandavkar

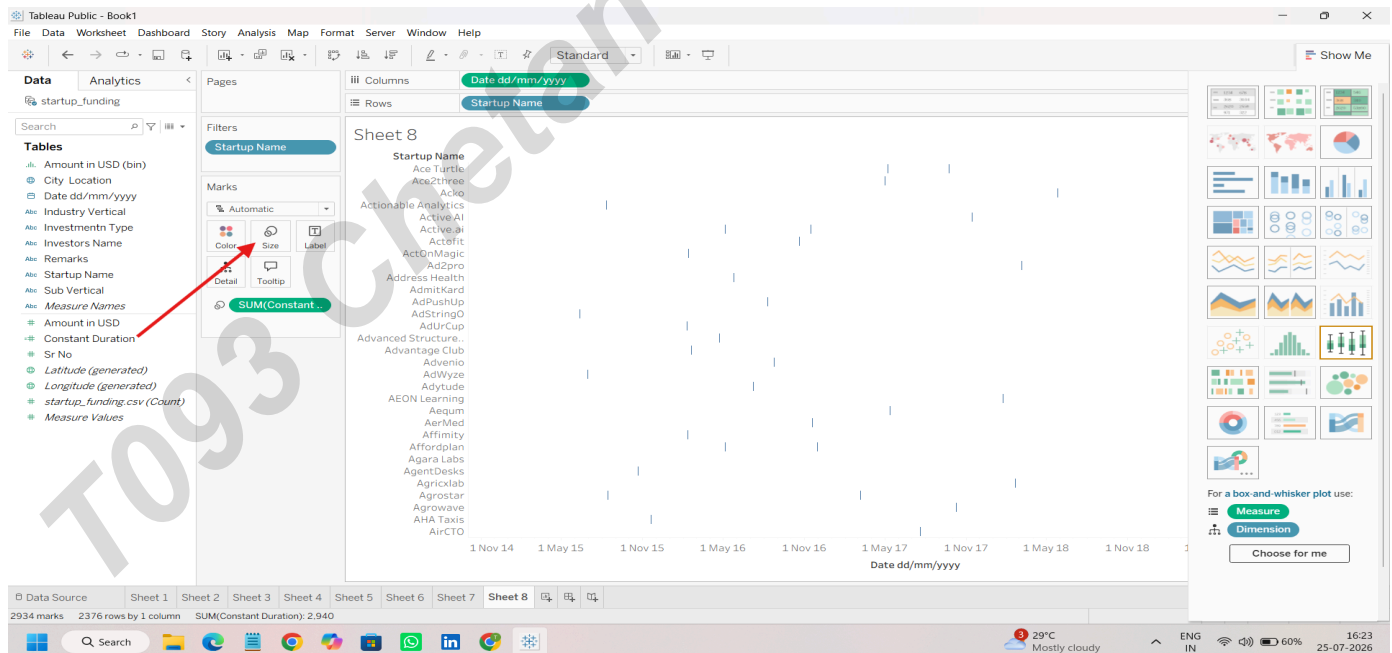
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Step 7:- Take your newly created calculated field Constant Duration from the Data Pane and drag it directly onto the Size tile inside your Marks card.



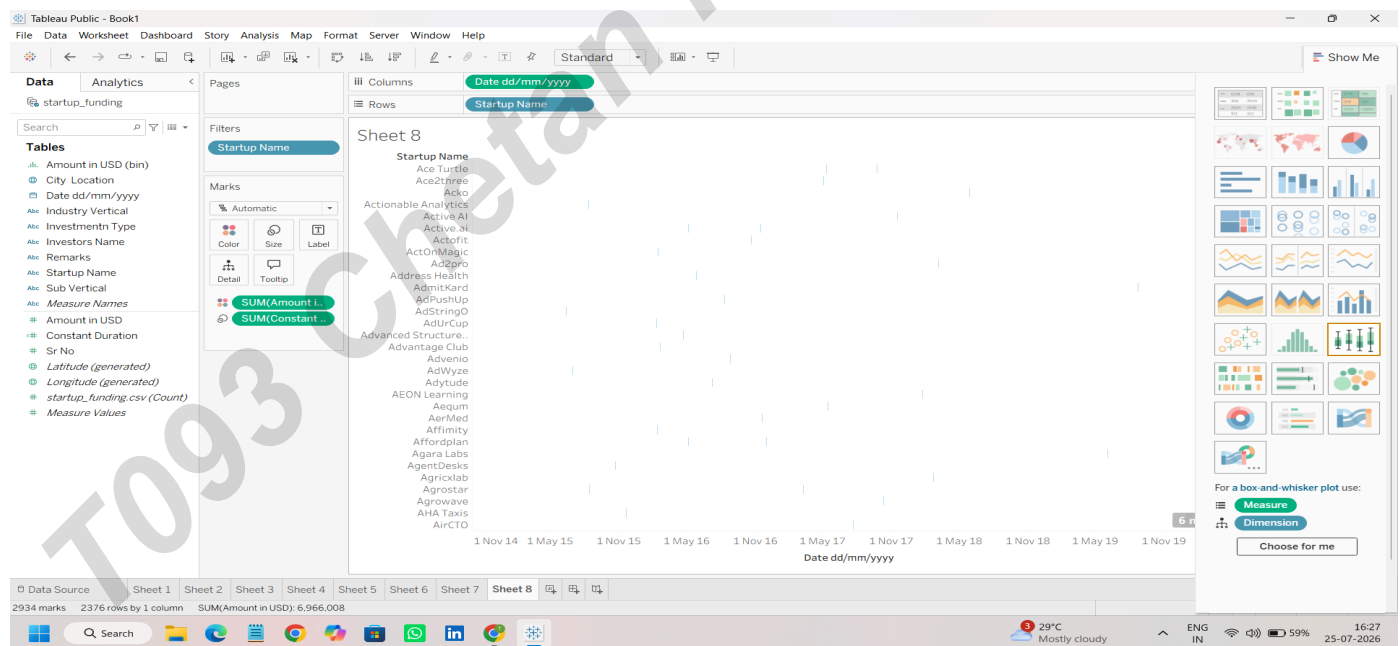
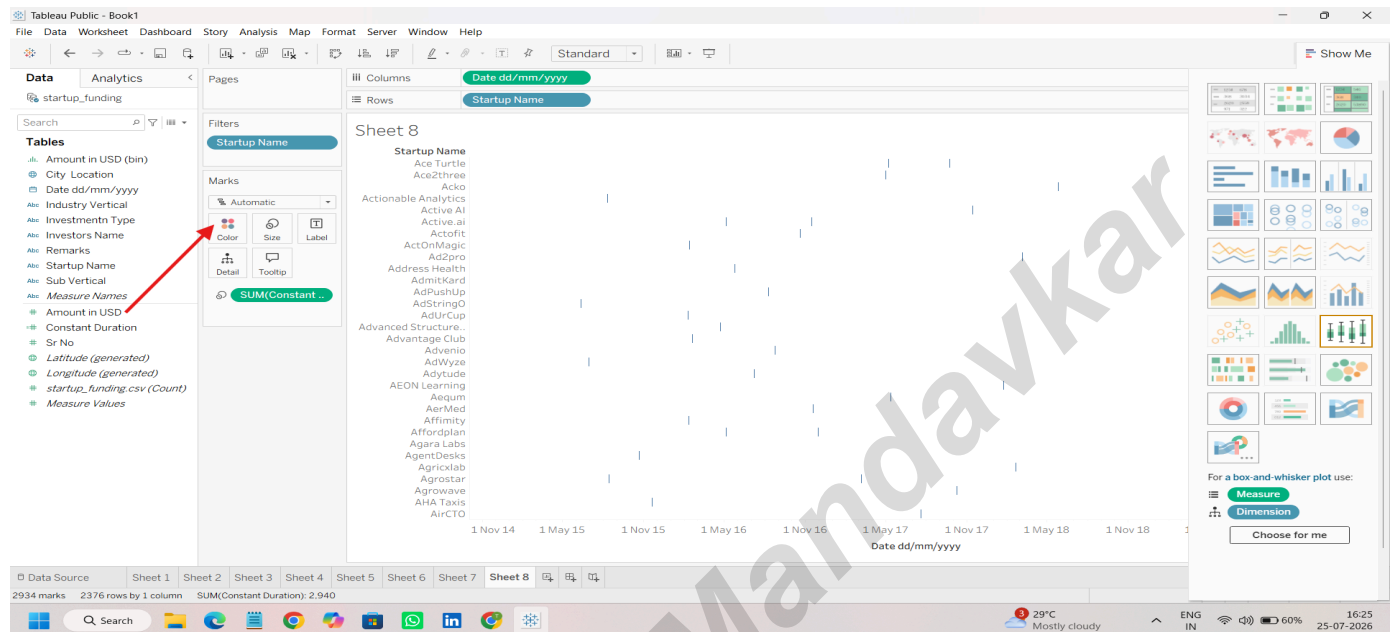
Name:- Chetan Mandavkar

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Step 8:- To make the timeline highly informative, drag your quantitative measure Amount in USD from the Data Pane and drop it directly onto the Color tile in the Marks card.



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